

Correlated Demand and Beta Dynamics in Sustainable Investing

YI FAN*

September 2025

ABSTRACT

Using U.S. active equity fund data from 2012 to 2024, this paper shows that correlated demand from sustainable investors amplifies systematic risk for stocks in their portfolios. I identify stocks persistently overweighted by sustainable investors as experiencing demand shocks. These stocks exhibit significant return beta increases beyond what fundamental characteristics would predict, with effects concentrated during 2016-2020. I develop a novel measurement capturing the coordination of capital flows across sustainable investors. This synchronized demand explains the systematic risk increases, while conventional investor demand shows no such coordination. These findings reveal a demand-driven channel through which thematic investing reshapes return betas.

*PhD student, Department of Finance, Stockholm School of Economics. Email: yi.fan@hhs.se

I. Introduction

The capital flows from sustainable investors have altered the landscape of investment over the past decade. Mutual funds and pension funds combined held ESG (Environmental, Social, and Governance) assets over 20% of global assets under management and are expected to reach \$40 trillion by 2030.¹ This expansion in thematic portfolio management leads to prominent **aggregate demand shifts** in the market, as investors tilt implicitly or directly to the ESG assets from the conventional investment. Previous literature has extensively discussed the financial performance of sustainable investing; however, the risk profile of the underlying securities is less understood. An important question in evaluating sustainable investing arises: Does securities' systematic risk exposure change due to the growing demand for sustainability?

I document a correlated demand channel that amplifies systematic risk in sustainable investing.² Examining the demand channel is economically important for two reasons. First, the structural foundation for correlated demand in sustainable investing is both distinctive and non-trivial. Sustainable funds exhibit pervasive portfolio overlap due to similar ESG criteria and sustainability mandates. This convergence in investment focus creates a concentrated subset of stocks that become disproportionately exposed to synchronized capital flows. It is further strengthened by multiple funds simultaneously responding to the same ESG sentiment signals, such as environmental events and regulatory changes. Strategic competition among fund managers also encourages green tilting or peer benchmarking (Basak and Pavlova 2013; Dumitrescu and Gil-Bazo 2024), leading to similarity in portfolio holdings even among funds without explicit ESG mandates. Second, understanding this demand channel helps disentangle the economic forces behind price appreciation in sustainable investing. If coordinated capital flows create systematic risk changes independent of fundamentals, it suggests that observed price movements may reflect demand-driven comovement rather than improvements in underlying prospects.

This paper focuses on return beta to assess systematic risk exposure. My methodology proceeds in several steps. I first identify the demand shocks in sustainable investing. Using

¹See <https://www.bloomberg.com/company/press/global-esg-assets-predicted-to-hit-40-trillion-by-2030-despite-challenging-environment-forecasts-bloomberg-intelligence/>

²This channel builds on Barberis and Shleifer (2003)'s asset class effect, where coordinated trading by category-based investors amplifies return co-movement beyond fundamentals.

a comprehensive dataset of U.S. domestic active equity mutual funds from 2012 to 2024, I construct the sustainable investing universe based on the aggregate holdings of ESG-focused funds from their prospectus-level strategy disclosures. This text-based approach captures funds that explicitly adopt sustainability criteria in investment decisions. Using detailed portfolio allocations and stock returns, I track stocks that experience increased capital flows from sustainable investors when they first become overweight in the sustainable investing universe and maintain this status for at least 12 quarters. I define it as an inclusion event in the representative sustainable investing portfolio. Importantly, conventional investors do not simultaneously increase their positions in these stocks. By examining actual portfolio holdings, I can directly observe when sustainable funds collectively increase their allocations to specific stocks, ensuring identification of precise demand shocks in each quarter rather than relying on fuzzy inference. Following Barberis, Shleifer, and Wurgler (2005), I measure systematic risk exposure through return beta coefficients, which come from the regression of individual stock returns on the representative sustainable investing portfolio returns. My identification strategy examines beta dynamics around these inclusion events and compares the affected stocks to observably similar controls matched on fundamental characteristics (Chen, Singal, and Whitelaw 2016).

One key finding is that event stocks' return betas with respect to the sustainable portfolio rise significantly compared to control stocks with similar fundamentals. The beta coefficients increase sharply after inclusion and remain elevated in the following three years. A decomposition of returns shows that stocks become more volatile and exhibit higher cross-correlations within the treated group. This additional systematic risk stems from exposure to coordinated demand shocks specific to the sustainable investing universe. This pattern reflects demand-induced risk, where changes in investor clientele mechanically alter stocks' beta exposures to common factors.

To quantify correlated demand, I measure each stock's exposure to coordinated sustainable investing demand shocks through regressions of individual stock flow-induced trading on aggregate sustainable portfolio flow-induced trading. The coefficient reflects how investors proportionally allocate capital flows between event stocks and the given sustainable portfolio, capturing coordination in the investor behavior. The sustainable funds demand coefficient explains contemporaneous return beta cross-sectionally in the three years following the inclusion. Conventional funds show little coordination after inclusion events. Moreover,

conventional investor demand coordination has no significant relationship with event stock return betas, indicating that the coordination measure captures sustainable investing-specific factors rather than generic market dynamics. It highlights sustainable investment’s distinctive feature: it reallocates not just capital but also systematic risk exposure. These findings reveal a novel transmission mechanism by which correlated capital flows from sustainable investors reshape individual stocks’ systematic risk exposure.

Related Literature. This paper relates to a broad literature of asset pricing in sustainable investing. Pástor, Stambaugh, and Taylor (2021), Pástor, Stambaugh, and Taylor (2022), and Pedersen, Fitzgibbons, and Pomorski (2021) develop the theoretical foundation of pricing sustainable assets. Recent empirical studies focus on the impact of aggregate sustainable investing demand shocks on asset prices (Van der Beck 2021), institutional investors’ profits (Koijen, Richmond, and Yogo 2023), and the real effects on the firms (Noh, Oh, and Song 2023). Other studies also linked sustainable investing to firms’ valuation (Giese et al. 2019; Albuquerque, Koskinen, and Zhang 2019); mutual funds’ performance (Geczy, Stambaugh, and Levin 2021); fund manager skills (Ceccarelli et al. 2023; Martí-Ballester 2015); ESG information, and fund financial merits (Birk, Jacob, and Wilkens 2024; Ciciretti, Dalò, and Ferri 2021; Cao et al. 2023; Rzeźnik, Hanley, and Pelizzon 2022); other institutional behavior (Pastor, Stambaugh, and Taylor 2024; Jeffers, Lyu, and Posenau 2024). I shift the focus from capital allocation and performance in sustainable investing funds to the stock beta dynamics. I highlight that additional risk emerges from correlated demand (Kim 2025; Han, Roussanov, and Ruan 2021), providing new evidence in the sustainable investing context. I shift the focus from capital allocation and performance in sustainable investing funds to stock beta dynamics. I highlight that additional risk emerges from correlated demand (Kim 2025), providing new evidence in the sustainable investing context.

This paper connects to the asset pricing literature on how coordinated investor behavior creates excessive return co-movement. Literature has documented return co-variation strengthened in index reconstitution (Barberis, Shleifer, and Wurgler 2005), stock splits (Green and Hwang 2009), and stocks with common institutional ownership (Anton and Polk 2014). These studies provide evidence through different correlated demand channels, such as investors caring about relative performance (Basak and Pavlova 2013) or institutional herding behavior (Wahal and Wang 2011; Dorn, Huberman, and Sengmueller 2008). My analysis extends this strand of literature to a new context where flows from sustainability-

oriented investors generate asset class effects and reshape the risk structure of stocks. *One key difference between my work and theirs is that I quantitatively examine the inelasticity in risk exposure by instrumenting the correlated demand from the flow-driven trading.* I develop a novel measurement of correlated demand that captures time-varying coordination patterns across different investor groups, enabling precise identification of when and how synchronized capital flows reshape systematic risk.

This paper also adds to the emerging literature on market efficiency and information transmission in ESG investment (D. Green and Roth 2021; Avramov et al. 2022; Giglio et al. 2025). By demonstrating that systematic risk changes occur independently of fundamental improvements, the findings suggest that sustainable investing growth creates new sources of undiversified risk. This has broad implications for understanding how thematic investing and sentiment-driven capital flows affect market structure and pricing efficiency (Giese et al. 2019; Starks 2023; Edmans, Gosling, and Jenter 2024).

The rest of the paper is structured as follows: Section II describes the construction of the sustainable investing universe and the stylized facts. Section III investigates the movements and structure of stock returns in a sustainable investing portfolio. Section IV explores the mechanism by which correlated demand imposes additional common risk exposure on related stocks. Finally, Section V presents the conclusions.

II. Sustainable Investing Universe

A. Data Collection

This paper employs a comprehensive dataset combining the regulatory filings, fund prospectuses, and financial market data from 2011 to 2024. The Center for Research in Security Prices (CRSP) Survivor-Bias-Free US Mutual Fund Database provides comprehensive data on mutual fund holding allocations and fund performance. I first matched the CRSP data with the Mutual Fund Prospectus Risk/Return Summary Data Sets from the Securities and Exchange Commission (SEC) website to obtain the disclosure information. After that, I identified all sustainable mutual funds through systematic textual analysis of fund prospectuses, focusing on whether they explicitly mentioned sustainability-related focus in their investment strategies. Stock-level data, including returns, prices, trading volumes, and market capitalization, are collected from the CRSP Stock and Index Database to assess the market impact of sustainable investment demand shocks. The appendix provides detailed definitions for all variables and construction methodologies.

B. Construction of the Sustainable Investing Universe

This screening process resulted in a total of 982 distinct active sustainable investing funds over the study period, with the number growing from 46 funds in 2011 to 647 funds by 2024 (out of 1874 active mutual funds, excluding dead funds) in the US market, reflecting the growth of sustainable mutual funds and the rapid expansion of the sustainable investing industry.³ To ensure comparability and focus on equity market impacts, the initial sample of sustainable funds was filtered to include only U.S. domestic equity funds. This filtering was performed using Lipper Objective Codes, a standardized classification system that categorizes mutual funds based on their stated investment objectives. The identified sustainable funds constitute the representative sustainable investor focusing on U.S. equities. The aggregate portfolio of sustainable mutual funds builds up *sustainable investing universe* (abbreviated as SI universe). In each quarter, the universe includes all stocks held by at least one sustainable fund in the sample, evolving from 788 unique stocks in 2012 Q1 to 3,219 stocks by 2024 Q2.⁴

³See Figure C1

⁴The full panel of sustainable investing universe is present in Table A1

Figure 1 plots the remarkable expansion of the SI universe market size from 2012 to 2024, revealing three distinct phases of growth. Sustainable investing assets under management remained modest before 2016, representing a small fraction of total market capitalization. Beginning in 2016, the sector experienced explosive growth, with total market value expanding dramatically through 2020 as both new funds entered the market and existing funds attracted substantial capital inflows. The expansion moderates after 2021, coinciding with the documented period of ESG skepticism and market reassessment.

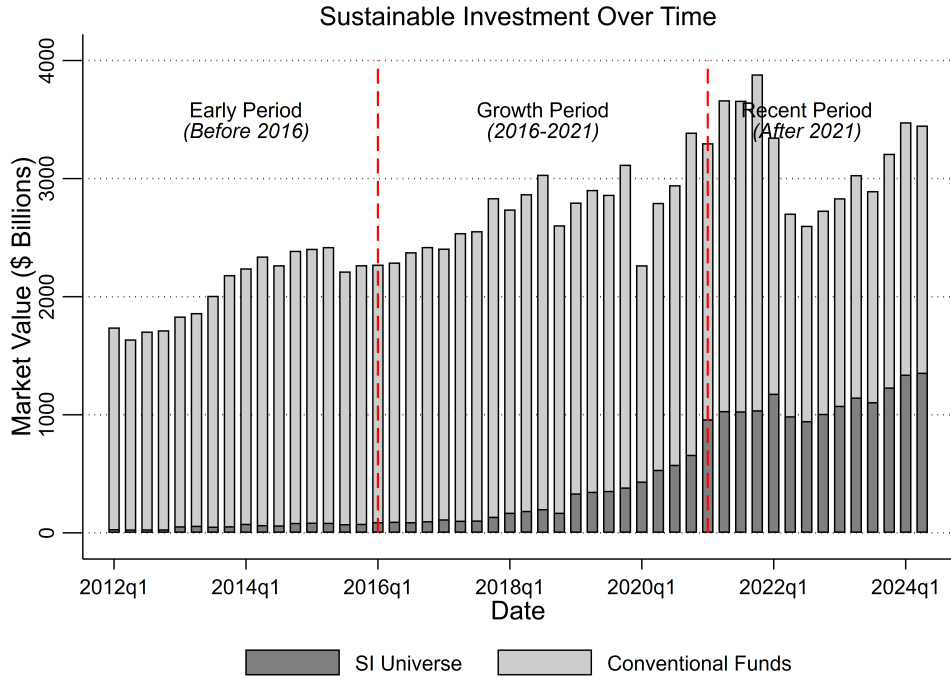


Figure 1. Growth of the Sustainable Investing Universe, 2012-2024. The bar graph shows the market size of identified actively managed sustainable mutual funds and the conventional mutual funds in the US domestic market, while the red boundaries show the structural change of the sustainable investing industry.

C. SI portfolio

I computed the sustainable investing weight (SI weight) as the share of stock market capitalization owned by sustainable funds relative to the total market capitalization of the SI

universe for each security constituting the SI universe. The active SI weight is the deviation of the SI weight from the stocks’ market portfolio weights.⁵ This relative weight measure helps identify stocks that are overweight in sustainable portfolios given the market benchmark. This measure captures the revealed preferences of sustainable investors, identifying which stocks they view as more attractive for ESG-focused portfolios.⁶ I define stocks with positive active weight in the SI universe as *SI stocks* from the perspective of sustainable investors. The *SI portfolio* is constructed based on these stocks preferred by sustainable investors, weighted by their market capitalization quarterly. Table I exhibits performance characteristics of the SI portfolio. There is a strong correlation between the monthly returns of the SI portfolio and the broader market. The correlation coefficient reaches 0.96 on the CRSP value-weighted index. Similarly, the correlation between the SI portfolio and the S&P 500 index stands at 0.95.

This finding challenges the common perception that sustainable investing represents a significant deviation from conventional market exposure. However, since the stocks preferred by sustainable investors have significantly proliferated, the high correlation is possibly driven by the close relationship with macroeconomic conditions and the growth of the SI industry in the time series. Therefore, it is essential to select the proper benchmark and determine the estimation periods when conducting the comparative analysis of stock returns.

D. Identify the Demand Shock

In this subsection, I propose a procedure to identify the sustainable demand shocks in the U.S. equity funds market. I track the SI weight for each stock in the universe and define *inclusion events* as a stock being first overweighted in the SI portfolio and having a consecutive holding record of at least 12 quarters following the inclusion quarter. This extended holding requirement serves two methodological purposes: (1) it filters out stocks that may be temporarily held for reasons unrelated to sustainability, and (2) it ensures enough sample to investigate the evolution and impact after the initial demand shock. This conservative approach provides the lower bound of the events of demand shocks.

This idea of inclusion events is inspired by index reconstruction and leverages the revealed preferences of sustainable investors. It captures actual portfolio allocation decisions of ESG-

⁵Based on CRSP value-weighted market portfolio.

⁶Similar identification is widely used in literature, like Van der Beck (2021).

Table I
Correlation Matrix and Average Monthly Returns

This table presents the correlation matrix and average monthly returns for seven indices over 153 monthly observations. The SI portfolio return is defined as:

$$R_t^{SI} = \sum_j s_{j,t}^{SI} r_{j,t}$$

where $s_{j,t}^{SI} = \frac{\theta_{j,t}}{\sum_{j \in SI} \theta_{j,t}}$ for R^{SI} (VW), the value weight of SI stock j based on market capitalization $\theta_{j,t}$. And $s_{j,t}^{SI} = 1$ for equally weighted R^{SI} (EW). R^{CRSP} (VW) is the Center for Research in Security Prices' value-weighted index covering the broad U.S. market. R^{CRSP} (EW) is the equally-weighted version of the CRSP index, where each stock receives the same weight regardless of size. S&P 500 tracks 500 large U.S. companies weighted by market capitalization. Returns are expressed as monthly percentages. R^{Non-SI} (VW) is computed as the size-adjusted (CAP) difference between the CRSP VW and the R^{SI} (VW), reflecting the aggregate performance of non-SI securities.

$$R_t^{Non-SI}(VW) = \frac{CAP_{t-1}^{CRSP}}{CAP_{t-1}^{CRSP} - CAP_{t-1}^{SI}} [R_t^{CRSP}(VW) - \frac{CAP_{t-1}^{SI}}{CAP_{t-1}^{CRSP}} R_t^{SI}(VW)]$$

R^{Non-SI} (EW) refers to the simple average return across all non-SI securities in the market. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Correlation Matrix							Average
	R^{SI} (VW)	R^{SI} (EW)	R^{CRSP} (VW)	R^{CRSP} (EW)	$R^{S\&P500}$	R^{Non-SI} (EW)	R^{Non-SI} (VW)	Returns (%)
R^{SI} (VW)	1							0.0142
R^{SI} (EW)	0.889***	1						0.0095
R^{CRSP} (VW)	0.967***	0.901***	1					0.0100
R^{CRSP} (EW)	0.896***	0.998***	0.914***	1				0.0078
$R^{S\&P500}$	0.955***	0.848***	0.992***	0.863***	1			0.0115
R^{Non-SI} (EW)	0.867***	0.998***	0.883***	0.995***	0.828***	1		0.0089
R^{Non-SI} (VW)	0.808***	0.785***	0.823***	0.798***	0.806***	0.772***	1	0.0064

focused funds. The surging portfolio weight in the SI universe indicates a potential demand shock from the perceived sustainable investors, where the transition from negative to positive active weights represents a clear signal that sustainable investors collectively treat this asset as a core investment. Unlike approaches that rely on fuzzy inference from third-party ESG index updates, this method directly observes investment decisions.

This procedure yields the 1,406 inclusion events between 2012Q1 and 2023Q2. For stocks added after 2021Q2, the 12-quarter holding requirement cannot be fully satisfied due to limited sample length. Nevertheless, I include them in the list since they are never underweighted once added. This identification of demand shock is robust using various holding windows, since once the stock starts to be overweighted in the SI universe, it rarely drops out.⁷ 185 stocks (13% of inclusion events) eventually transition from overweight back to underweight status after maintaining positive active weights for twelve quarters. The following analysis focuses on demand shocks in inclusion because it is more interesting and prominent.

Alternatively, I identify the persistent demand shocks as stocks having consecutive positive active weights for 12 quarters and define the first quarter as the inclusion quarter. This more restrictive approach yields a smaller set of 874 stocks and a slightly larger demand shock, but produces similar return beta dynamics shown in the appendix, confirming that the main results are not sensitive to the specific definition of persistence.

⁷As a robustness check, I vary the persistence requirement for defining inclusion events. Using thresholds of 8 and 16 quarters, instead of the baseline 12 quarters, results in 1,464 and 1,370 event stocks, respectively. The differences are mainly driven by post-2020 samples. Since most demand shocks are concentrated in the period before 2020, the results are not sensitive to the holding length.

III. Sustainable Demand Shock and Return Beta

A. *SI Weight Changes Around Inclusion Events*

Figure 2 reveals the dynamic changes in SI weight for each inclusion event in the sustainable investing universe. Before the inclusion, stocks were underweight or had missing holding data, as their active portfolio weight in the SI universe is negative. From the inclusion quarter, the weight jumped to a significantly large level and remained. I plot the average active weight in the conventional active equity mutual funds in the dashed line, showing a very minimal variation.⁸ However, considering the huge difference between the market size of the SI universe and the market universe before 2016, I supplement the quantity changes measured by the share numbers, as shown in Figure 3.

The demand shocks exhibit both sharp initial increases and persistent patterns. As illustrated in Figure 2, the average active portfolio weight jumps from negative territory to approximately 0.02% in the inclusion quarter and stabilizes at elevated levels thereafter. The increased equity demand from sustainable investors represents approximately 2.5% of outstanding shares on average, underscoring the economic significance of these demand shifts.

B. *Price Effects*

This subsection analyzes the price effect of sustainable investing demand shocks in the short term. Specifically, I construct cumulative abnormal return trajectories spanning different periods for each event stock. As illustrated in Figure 4, three distinct lines represent inclusion events that occurred before 2016, between 2016 and 2020, and 2021 onward. Each trajectory tracks the cumulative abnormal returns (calculated as the raw daily returns minus market returns) over a window extending from 120 days pre-inclusion through 150 days post-inclusion. The results reveal notable temporal variations in the price response patterns. Demand shocks that occurred before 2020 demonstrate broadly similar price effects, though with different magnitudes. The sample in the 2016-2020 period exhibits more accelerated cumulative return growth compared to those included before 2016. However, a marked shift occurs for inclusions from 2021 onward, where cumulative returns follow a distinctly different pattern, peaking around the end of the inclusion quarter before entering a pronounced decline phase. This evolving response pattern suggests a potential structural change in how

⁸The conventional investor is defined as an active equity fund without a specific ESG focus.

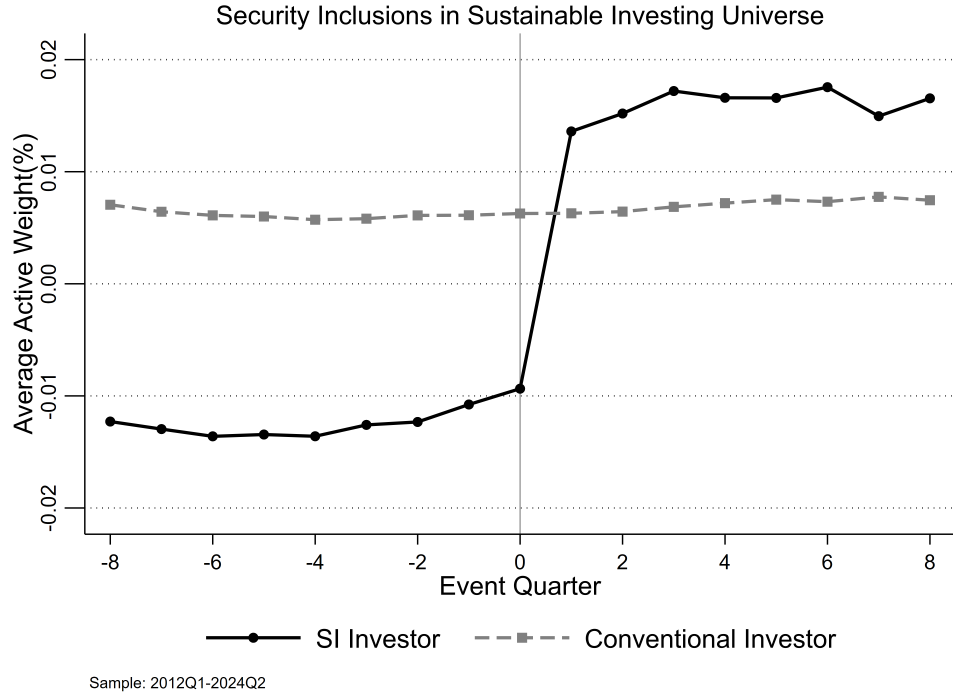


Figure 2. Active Sustainable Investing Weight of Event Stocks. This figure shows the average active positions in sustainable and conventional funds for event stocks from 2012Q1 to 2024Q2. The x-axis represents quarters relative to the inclusion event. The black line shows active positions by sustainable investors, where weights in the SI universe exceed market portfolio weights. The gray dashed line represents conventional investors, defined as US active equity mutual funds that do not follow specific ESG investment strategies. The sharp and persistent rise in the SI universe highlights the demand shock from sustainable investors following inclusion events.

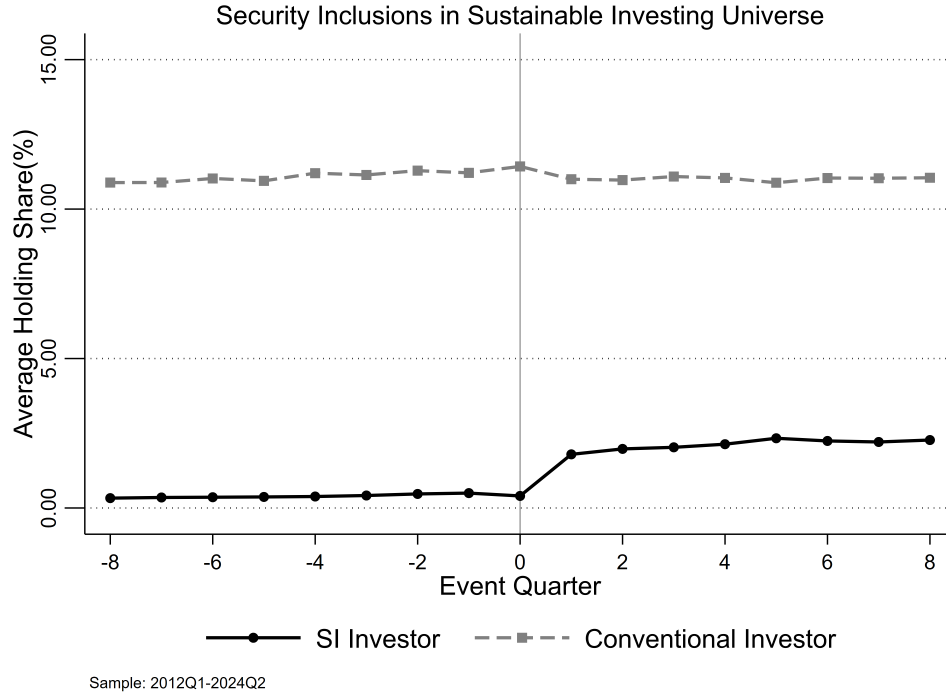


Figure 3. Sustainable Investing Equity Demand of Event Stocks. This figure illustrates equity ownership patterns around stock inclusion events in the sustainable investing universe from 2012Q1 to 2024Q2. The x-axis represents quarters relative to the inclusion event. The black line shows the ownership share by sustainable investors, measured as the proportion of total outstanding shares held by these funds. The gray dashed line shows the corresponding ownership share by conventional investors, defined as US active equity mutual funds that do not follow specific ESG investment strategies.

markets process and subsequently reassess sustainability-related demand shocks in more recent periods. Although price impact is not the focus of this paper, the slope of cumulative returns of event stocks temporarily reflects price movement characteristics.

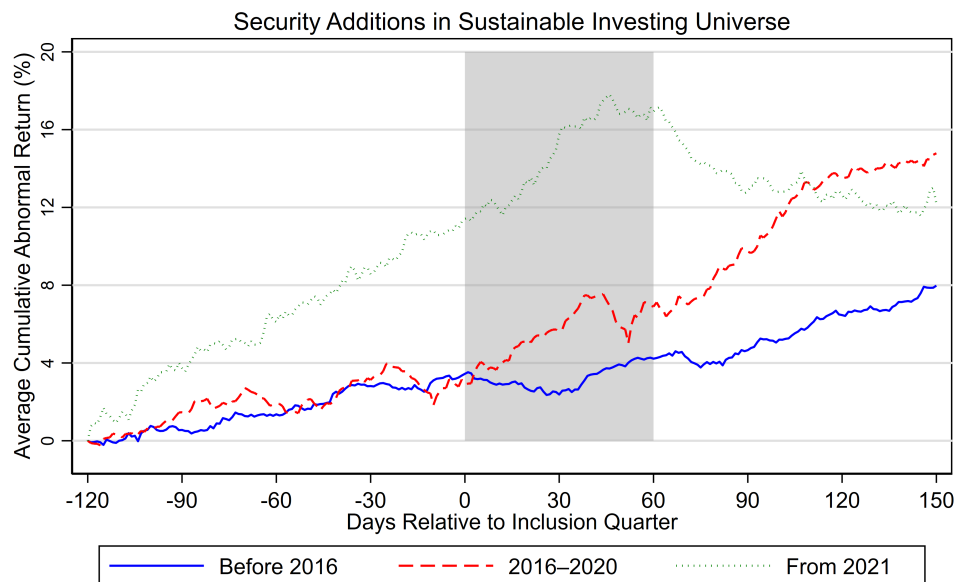


Figure 4. Cumulative Abnormal Returns of Event Stocks. The figure presents the cumulative abnormal returns for inclusion events that occurred during different periods: before 2016 (blue line), between 2016 and 2020 (red line), and 2021 onward (green line). The shaded gray area indicates the inclusion quarter. Cumulative abnormal returns are calculated by subtracting the market return from the raw daily return of each stock. The analysis spans 120 days before the first day of the event quarter through 150 days after.

C. Beta Changes

Return beta quantitatively measures how individual stock prices covary with another asset. This subsection investigates how demand shocks from sustainable investors affect the stock price movements by estimating the price covariance between the event stocks and the SI portfolio.

Following Barberis, Shleifer, and Wurgler (2005), the return beta of a given portfolio is shown in the following single-factor model,

$$r_{i,t} = \alpha_i + \beta_i^p R_t^p + u_{i,t} \quad (1)$$

where $r_{i,t}$ represents the return of stock i from t to $t + 1$, R_t^p is the return of a given portfolio, and β_i^p captures the stock's sensitivity to the given portfolio returns. I estimate sustainable investing betas (β_i^{SI}) using equation (1) for each stock in the inclusion event. This estimation is run over two distinct periods: the pre-inclusion period (12 months before the inclusion quarter) and the post-inclusion period (12 months after the inclusion quarter). The benchmark portfolio includes all SI stocks that have existed in the sustainable investing universe for at least one year at quarter t .⁹

The interpretation of the regression coefficient is straightforward. A higher SI beta shows enhanced comovement between individual stocks and the SI portfolio. The economic understanding is that it reflects the sensitivity to the risk factors that are shared by the SI portfolio. Therefore, the change in beta ($\Delta\beta$) is calculated as $\beta_{post} - \beta_{pre}$ to capture the shift in risk exposure for each stock following inclusion. Such beta shifts reveal innovations in fundamental factors or behavioral elements.

Table II reports beta raw changes in daily, weekly, and monthly frequency, where the weekly and monthly beta have an extended window of three years.

As shown in the table, there is a clear temporal pattern in different periods. During the early phase of sustainable investing, newly included stocks show negative beta changes (significant on a daily basis), suggesting a limited connection with the market movement. However, from 2016 to 2020, a period marked by accelerated SI growth, the delta beta

⁹In other words, I restrict event stocks to the period 2013Q1-2023Q2 (1170 stocks). This restriction allows us to estimate pre-inclusion one-year rolling betas using a consistent benchmark portfolio, as our SI portfolio construction begins in 2012Q1. This approach ensures that event stocks have a full one-year window of pre-inclusion data for beta estimation without forward-looking bias.

Table II
Change in SI Beta by Period and Frequency

This table presents changes in beta relative to the sustainable investing (SI) portfolio at daily, weekly, and monthly frequencies. The SI portfolio return is defined as:

$$R_t^{SI} = \sum_j s_{j,t}^{SI} r_{j,t}$$

where $s_{j,t}^{SI} = \frac{\theta_{j,t}}{\sum_{j \in SI} \theta_{j,t}}$ is the value weight of SI stock j based on market capitalization $\theta_{j,t}$. The beta coefficient is estimated as:

$$\beta_t^{SI} = \frac{\text{cov}_t(r_{i,t}, R_t^{SI})}{\text{var}_t(R_t^{SI})}$$

$\Delta\beta = \beta_{post} - \beta_{pre}$ measures the difference between pre- and post-inclusion betas. For daily and weekly frequencies (Panels A and B), estimation periods span 12 months before and after the inclusion quarter: [-12,-1] and [1,12]. For the monthly frequency (Panel C), estimation periods expand to 36 months. The sample is divided into three sub-periods: 2013–2015, 2016–2019, and 2020–2023. Stocks with less than 80% of available trading days during the estimation window are excluded. Standard errors are reported in the final column.

Panel A: Daily SI Beta			
Period	Observations	$\Delta\beta^{SI}$	Std Err
2013–2015	336	-0.0360	0.0178
2016–2019	375	0.0785	0.0261
2020–2023	480	0.0257	0.0230
Panel B: Weekly SI Beta			
Period	Observations	$\Delta\beta^{SI}$	Std Err
2013–2015	336	-0.0173	0.0301
2016–2019	372	0.0948	0.0433
2020–2023	477	-0.0203	0.0421
Panel C: Monthly SI Beta			
Period	Observations	$\Delta\beta^{SI}$	Std Err
2013–2015	304	-0.1320	0.0387
2016–2019	305	0.1246	0.0472
After 2020	351	-0.1171	0.0392

shows consistent positive beta changes across all specifications, with statistically significant increases in risk exposure. The magnitude of these changes increases with the measurement horizon, from approximately 0.08 at daily frequency to 0.12 at monthly frequency. After 2021, a mixed pattern emerges in the most recent period, with positive but diminishing beta changes at the daily frequency, and significantly negative changes at the monthly frequency.

The evidence suggests a temporal pattern where increased sustainable investing sentiment in the market coincides with enhanced price co-movement among stocks with SI membership. I check the market beta evolutions of the same sample using a CAPM model, and find similar beta shifts reported in the Appendix Table [A2](#). Altogether, the price covariance changed significantly, reflecting the innovations to systematic risk.

IV. Mechanisms Behind Increased Return Beta

How to understand the return beta of stocks increasing around the event shown in Section III? One might think that stocks preferred by sustainable investors are similar in terms of fundamentals; for example, event stocks are supposed to have a larger cash flow in the future. Thus, the increased beta is attributed to the realized higher cash flow. Alternatively, Barberis and Shleifer (2003) propose the asset class effect, which is created by correlated demand that is even unrelated to fundamentals. As Barberis, Shleifer, and Wurgler (2005) show, a stock has been added to the S&P 500 move with other securities because benchmark-linked investors who own the stock also own the other securities in the portfolio.

In this study, I investigate the distinctive scenario where stocks in the SI universe exhibit increased co-movement due to shared ownership by ESG-focused institutional investors. I consider a new channel of mechanism from the inelastic demand in sustainable investing, where institutional investors unconsciously buy or sell assets that go beyond reactions to individual stock fundamentals. The channel was first mentioned in Greenwood and Thesmar (2011), who presented that the correlated capital flows from the liquidity needs of institutional investors predict the price co-movement. In the settings of sustainable investing, correlated capital flows into a security occur because of increased exposure to the shared owners.

This section involves examining the mechanism through which correlated demand shocks in sustainable investing affect stock price dynamics. I first show the non-fundamental systematic risk tangibly by presenting the changes in the stock's factor loadings that cannot fully cover the changes in return beta. Then, I develop the measurement of correlated demand in SI funds and test its predictions for the increased return betas after the inclusion.

A. Beyond Traditional Comovement Mechanisms

The first step is to reexamine how much fundamental factors account for the variations in the risk change. The following two charts in figure 5 indicate that SI funds tend to favor larger stocks and those with past returns above the median. However, there is no significant strictly positive relationship with these trends. Next, I check the fundamental growth rate around the quarter of inclusion. I found no significant evidence showing that there are fundamental improvements in figure 6.

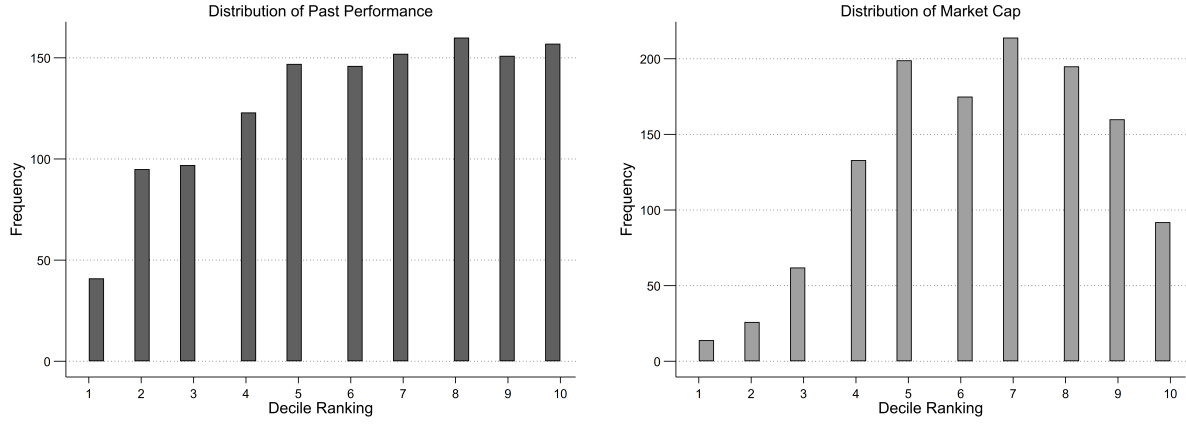


Figure 5. Descriptive Statistics of Event Stock Returns and Performance. This figure presents the distributional characteristics of stocks entering the sustainable investing universe. The left panel shows the frequency distribution of event stocks across cumulative return deciles over the past year, measured at the time of inclusion. The right panel displays the frequency distribution across market capitalization deciles. Both distributions demonstrate that SI funds exhibit preferences for larger firms and past winners, though without strict monotonic relationships.

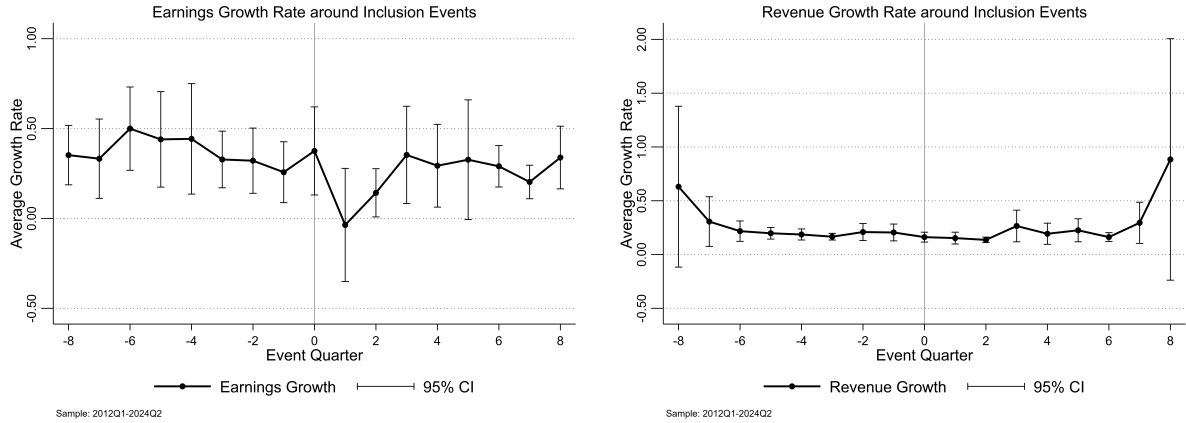


Figure 6. Fundamental Growth Rate around Events This figure displays the average earnings growth rate (left panel) and revenue growth rate (right panel) for event stocks around their inclusion in the sustainable investing universe. The x-axis represents quarters relative to the inclusion event. Earnings growth is calculated as $(EPS_t - EPS_{t-4})/EPS_{t-4}$ for stocks with positive EPS_{t-4} . Revenue growth is calculated as $(Sales_t - Sales_{t-4})/Sales_{t-4}$. The absence of significant changes around the inclusion quarter suggests that sustainable investing demand shocks are not driven by fundamental performance improvements.

A.1. Matching Analysis with Control Groups

Institutional investors typically favor large companies and past winners. This observation raises concerns documented in Chen, Singal, and Whitelaw (2016), which argue that the price co-movement in the S&P 500 index can be largely explained by the endogenous evolution of stocks' fundamental factor loadings (Chen, Singal, and Whitelaw 2016; Kasch and Sarkar 2013). The key findings are that S&P 500 stocks are also high momentum stocks with declining momentum beta in the following year, resulting in increasing regression coefficients. When controlling for momentum in the matched stock group, there is no excess co-movement between index inclusion stocks and the existing portfolio.

To address this concern in our case, I conduct a one-on-one matching based on size, past return, and the closest market beta of event stocks to control the fundamental factor loading changes. The analysis in Table III examines whether beta changes observed around inclusion quarters also occur in matching stocks with similar fundamentals. When comparing stocks that match as a control group, the double difference analysis shows that the changes in beta specifically reflect the impact of sustainable investing membership rather than the mechanical result of evolving fundamental characteristics that affect systematic risk. On average, the change in beta of 0.0761 is in excess of untreated stocks during 2016 to 2020. One-year rolling beta of event stocks and matched stocks on the SI portfolio are presented in the Figure A2, showing a striking contrast between event stocks and their matched controls. Event stocks exhibited significant increases in SI beta following inclusion, with the 12-month rolling SI beta showing a clear upward trajectory starting around the inclusion quarter. Conversely, matched control stocks lack a similar trend in SI beta over the same period.

Table III
SI Beta Change Analysis with Matching Stocks by Time Period

This table presents a difference-in-differences analysis of beta changes relative to the sustainable investing (SI) portfolio. $\Delta\Delta\beta$ represents the difference-in-differences estimate calculated as: $\Delta\beta_{event}^{SI} - \Delta\beta_{matched}^{SI}$. $\Delta\beta_{event}^{SI}$ measures the change in SI portfolio beta for event stocks, and $\Delta\beta_{matched}^{SI}$ measures the corresponding change for matched control stocks. The SI portfolio beta is estimated as:

$$\beta_t^{SI} = \frac{\text{cov}_t(r_{i,t}, R_t^{SI})}{\text{var}_t(R_t^{SI})}$$

where R_t^{SI} is the market-cap weighted SI portfolio return. Control stocks are matched on market capitalization, past performance, and market beta. The analysis tests whether inclusion in the sustainable investing universe affects systematic risk exposure as measured by beta changes relative to the SI portfolio. Estimation periods span 12 months before and after the inclusion quarter: $[-12, -1]$ and $[1, 12]$. Stocks with less than 80% of available trading days during the estimation window are excluded. Robust t -statistics are reported in parentheses. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Before 2016			
	$\Delta\beta_{matched}^{SI}$	$\Delta\beta^{SI}$	$\Delta\Delta\beta$
	-0.0466** (-2.098)	-0.0367** (-2.021)	0.0206 (0.930)
Observations	283	331	283
Panel B: 2016–2020			
	$\Delta\beta_{matched}^{SI}$	$\Delta\beta^{SI}$	$\Delta\Delta\beta$
	-0.0054 (-0.237)	0.0612*** (2.690)	0.0761*** (2.886)
Observations	423	511	423
Panel C: From 2021			
	$\Delta\beta_{matched}^{SI}$	$\Delta\beta^{SI}$	$\Delta\Delta\beta$
	0.0468* (1.761)	0.0095 (0.348)	-0.0260 (-0.800)
Observations	234	311	234
Panel D: Full Sample			
	$\Delta\beta_{matched}^{SI}$	$\Delta\beta^{SI}$	$\Delta\Delta\beta$
	-0.00481 (-0.345)	0.0191 (1.411)	0.0340** (2.139)
Observations	940	1,153	940

A.2. A Model of Decomposing Return

To understand the excessive beta coefficient, this subsection explores the return dynamics through a factor model.

I first run the regression of individual stock returns on the classic factors, the market, the Fama-French three factors, and the five factors in Table IV. I computed the average R-squared of the fundamental factor regressions for event stocks and found that, on average, it decreased by -8% to -14% after the inclusion.

The decreasing explanatory power of the fundamental factor model indicates that a novel risk component has influenced asset prices. Let us decompose the stock return in the following ways:

$$r_{i,t} = f_t + e_{i,t} \text{ (Pre-inclusion)} \quad (2)$$

$$r_{i,t} = \eta_{i,t}f_t + \gamma_{i,t}\mu_t^{SI} + e_{i,t} \text{ (Post-inclusion)} \quad (3)$$

In this context, μ_t^{SI} denotes sustainable investing-specific group risk, which is unrelated to fundamentals $\mathbf{Cov}(\mu_t^{SI}, f_t) = 0$. The post-inclusion return implies that, from the event quarter, stock returns are driven by not only the common factor but also a potential risk from the aggregate sustainable investing demand μ_t^{SI} .

Under the assumption that idiosyncratic volatility σ_e and fundamental factor volatility σ_f remain constant in the short term around inclusion, the increasing fundamental factor loading $\eta_{i,t} > 1$ and decreasing R-squared reflect the information that $\gamma_{i,t} \neq 0$.

Intuitively, traditional factor models assume that systematic risk stems from the covariance of fundamental economic forces like market movements, size effects, or value premiums. When a non-fundamental risk shock emerges, such as coordinated demand from sustainable investors, it strengthens price covariance that operates independently of these fundamental correlations. This additional source of risk impairs the explanatory power of traditional models because it increases total volatility while exceeding the part explained by fundamental factors. Empirically, the decreasing R-squared and the increasing explained variation by f_t deliver this message exactly.

Given this framework, I interpret the increased return betas of event stocks as reflecting larger exposure $\gamma_{i,t}$ to a demand-driven risk factor specific to the SI portfolio. A concern is that time-varying σ_e may invalidate this argument. However, in Table A3, I show that

Table IV
Explanatory Power of Factor Models Before and After Inclusion

This table reports the average R-squared and root mean squared error (RMSE) from time-series regressions of event stock returns on various factor models from 2012 to 2024. For each stock, I estimate factor models using daily returns over 12-month windows before (Pre-Event) and after (Post-Event) the inclusion quarter. Stocks with less than 80% of available trading days during the estimation window were excluded from this table. Panel A reports results for the single-factor CAPM, Panel B for the Fama-French three-factor model, and Panel C for the Fama-French five-factor model. Column (3) reports the change in each statistic (Post minus Pre), and column (4) reports the percentage change.

	Pre-Event (1)	Post-Event (2)	Change (3)	% Change (4)
<i>Panel A: CAPM</i>				
Average R^2	0.3144	0.2698	-0.0446	-14.2%
Average RMSE	0.0193	0.0191	-0.0002	-1.0%
<i>Panel B: Fama-French Three-Factor Model</i>				
Average R^2	0.3762	0.3441	-0.0321	-8.5%
Average RMSE	0.0183	0.0180	-0.0003	-1.6%
<i>Panel C: Fama-French Five-Factor Model</i>				
Average R^2	0.3930	0.3628	-0.0302	-7.7%
Average RMSE	0.0181	0.0178	-0.0003	-1.7%
Observations	1052	1052		

event stocks' idiosyncratic risk cannot explain the beta change. In particular, I construct an event-quarter-specific portfolio using all other event stocks that share the same event quarter for each event stock and compute the correlation of 12-month daily returns between the individual stock and this peer portfolio before and after the event quarter. The results show that, upon entering the universe of sustainable investment, the correlations between event stocks increase, suggesting enhanced connections. It is clear that merely idiosyncratic risk cannot explain this pattern. While it is difficult to detect the group risk driven by sustainable demand shocks directly, in the next section, I develop a measurement of synchronized trading among investors to index the exposure to the demand shock. In other words, instead of measuring μ_t^{SI} , I measure $\gamma_{i,t}$.

B. The Correlated Demand Mechanism

The preceding analysis provides a conceptual example of how the shared demand shock adds additional price comovement for the stocks in the sustainable investing universe. In this section, I attempt to measure the shared demand shock by projecting capital flows from different investors. Intuitively, whenever investors buy(sell) one dollar of stock A, they simultaneously trade one dollar in the same direction for stock B; the capital flows received by the two stocks are 100% correlated. As a result, two stocks share 100% demand shock.

Unlike the shared ownership measurements used in Anton and Polk (2014), Grégoire (2020), and Kim (2025), where researchers measure the size of the aggregate flows of total investors to explain the variation of the asset price, I decompose the magnitude of coordinated trading and focus on the degree to which a stock attracts coordinated demand from different investors.

I first follow Greenwood and Thesmar (2011) to measure the magnitude of correlated dollar flows into security from the SI universe by regressions of flow-driven transactions of individual stock on the aggregate flow-driven trading of the SI portfolio across funds. Then, I show how the correlated flow patterns are different for sustainable-focused funds and conventional funds. Lastly, I empirically confirm the model prediction that stocks covary more with the SI universe as the correlated sustainable demand is enhanced.

B.1. Theoretical Foundation

Greenwood and Thesmar (2011) demonstrate that aggregate capital from flow-driven investors can predict the contemporaneous return of that security i ¹⁰:

$$r_{i,t+1} = \psi_{i,t} + \lambda_i \frac{\sum_k w_{i,k,t} f_{k,t}}{\theta_{i,t}} + \epsilon_{i,t+1} \quad (4)$$

where $r_{i,t+1}$ is the return of security i in the period from t to $t + 1$. $\theta_{i,t}$ is a scaling factor, usually the security's market capitalization. $w_{i,k,t}$ and $f_{k,t}$ refer to the portfolio weight of security i owned by the investor k , and the dollar flows into investor k in the period from t to $t + 1$. $\epsilon_{i,t+1}$ has a mean of zero conditional on the news received at time t .

$$\beta_{i,t}^{SI} = \frac{\text{cov}_t(r_{i,t}, \sum_j s_{j,t}^{SI} r_{j,t})}{\text{var}_t(\sum_j s_{j,t}^{SI} r_{j,t})} \quad (5)$$

Equation (5) shows the return beta of stock i on the SI portfolio, where $s_{j,t}^{SI} = \theta_{j,t} / \sum_{j \in SI} (\theta_{j,t})$ is the weight in SI portfolio for a security j based on their market capitalization at time t . By substituting individual stock returns with the received aggregate capital flows from equation (4), we obtain the co-fragility of the SI portfolio, $F_{i,t}$.

$$F_{i,t} = \frac{\text{cov}_t\left(\frac{\sum_k w_{i,k,t} f_{k,t}}{\theta_{i,t}}, \sum_j s_{j,t}^{SI} \frac{\sum_k w_{j,k,t} f_{k,t}}{\theta_{j,t}}\right)}{\text{var}_t\left(\sum_j s_{j,t}^{SI} \frac{\sum_k w_{j,k,t} f_{k,t}}{\theta_{j,t}}\right)} \quad (6)$$

Equations (4), (5), and (6) show that the correlation between asset return r_i and portfolio return can be linked to the correlation between the aggregate capital flows into each asset. In particular, if capital flows contribute to a 100% variation in asset prices, then flow beta should have a one-to-one relationship with return beta. Note that equation (6) can be decomposed into three components: the number of total investors, aggregate fund flow volatility, and the coordination across investors' trading. While the aggregate fund flow volatility is assumed to be exogenous, coordination across investors' trading—holding the number of investors fixed—determines the degree of correlated demand. I focus on the coordination trading pattern as it captures the unique group risk driven by a subset of investors, rather than aggregate flows from all market participants. This distinction is important because the

¹⁰See Appendix for the modeling links between the asset returns and fund flows.

previous analysis shows that while aggregate SI fund size continuously increased over the sample period, the price co-movement patterns only strengthened during 2016-2020. To understand this temporal pattern, it is necessary to isolate the demand shock from specific investor groups rather than relying on total market trends.¹¹

The following empirical design measures this coordination directly by estimating how individual stock flows respond to portfolio-wide flows within different fund groups (sustainable funds or conventional funds), providing a more targeted view of demand synchronization than aggregate co-fragility measures.

B.2. Measuring Demand Synchronization

Assume there is a representative SI investor that receives the average fund flows. Then the demand coordination can be approximated by equation (7).

In each quarter, $D_{i,t}$ represents the regression coefficient of the mechanical capital flows to security i on the capital flows to all other securities j in the current sustainable investing portfolio across equity funds k .¹²

$$\frac{w_{i,k,t}f_{k,t}}{\theta_{i,t}} = \alpha_{i,t} + D_{i,t} \frac{\sum_j w_{j,k,t}f_{k,t}}{\sum_j \theta_{j,t}} + u_{i,k,t+1} \quad (7)$$

The coefficient reflects how much normalized flow into stock i is responsive to total normalized flow into a given portfolio. Two main factors drive this process: representative fund flows and similarity in the fund portfolio weights. When investors hold the same portfolio with the fixed portfolio weight, the demand coefficient turns out to be the correlated capital inflows that the fund receives. When every investor experiences the same amount of capital inflows, the coefficient reflects the similarities in their ownership of stock i . Several scenarios help clarify the interpretation of $D_{i,t}$. A value of zero for $D_{i,t}$ indicates no linear relationship between investors' capital allocation to stock i and the referenced portfolio. A positive $D_{i,t}$ means that when funds receive inflows, they assign capital to stock i in proportion to allocations in the SI portfolio. In contrast, a negative $D_{i,t}$ implies that investors trade stock i

¹¹In the appendix, I show how co-fragility can be approximated as a linear function of cross-sectional coordination coefficients.

¹²To calculate the coefficient, we need at least three observations for each stock in every quarter. Therefore, I filter out stocks that have fewer than three owners in the sample data.

differently from the SI portfolio. Missing values of $D_{i,t}$ reflect investors holding zero positions in stock i or minimal investor participation.

B.3. Testing Framework and Predictions

I estimate separate coefficients $D_{i,t}^S$ and $D_{i,t}^N$ for sustainable and conventional funds, respectively, showing how coordination patterns differ across investor types. The model of flow-driven returns in equation (4) generates two key predictions for demand coordination. First, demand coordination estimated from the ESG-focused fund, $D_{i,t}^S$, should increase sharply after the stock i becomes a member of the SI portfolio. This increase reflects both the common capital flows from sustainable investors and the convergence of portfolio weights as stocks transition from being underweight to overweight.

Second, if coordinated demand drives price correlation, stocks with higher demand coordination should exhibit a larger return beta on the SI portfolio. To test this relationship, I estimate regression 8:

$$\beta_{i,t}^{SI} = a_t + b \times D_{i,t} + C'X_{i,t} + \varepsilon_{i,t} \quad (8)$$

where $\beta_{i,t}^{SI}$ is the systematic risk exposure of stock i to the SI portfolio at time t . $D_{i,t}$ represents the demand coordination measure for the relevant fund group (either $D_{i,t}^S$ for sustainable funds or $D_{i,t}^N$ for conventional funds). The vector $X_{i,t}$ includes control variables, and a_t represents time fixed effects that control for market-wide factors affecting all stocks in period t .

If the coefficient b is positive and statistically significant, then it can provide evidence for the correlated demand channel. I now turn to empirical analysis of demand coordination patterns and their effects on return beta around inclusion events.

C. Results

C.1. Demand Dynamics Around Inclusion

Table V presents summary statistics of demand coefficients and SI betas across three key periods: pre-inclusion, the first three years post-inclusion (Q1-Q12), and the longer-term period (Q13 onwards). The table reveals striking patterns in coordination behavior between sustainable and conventional investors.

Table V
Summary Statistics by Event Period

This table presents summary statistics for demand coordination coefficients and SI betas across three periods: before inclusion, quarters 1-12 after inclusion, and quarters 13+ after inclusion. The sample includes 1,406 event stocks from 2012Q1 to 2024Q2. D^N measures conventional fund demand coordination and D^S measures sustainable fund demand coordination, both estimated from equation (7). β^{SI} represents quarterly return betas with respect to the SI portfolio, using 60 trading days. To ensure reliable estimation, stocks with fewer than 3 fund owners of each type are excluded, yielding 1,369 stocks with conventional coordination measures and 1,347 stocks with sustainable coordination measures.

Variable	Pre-inclusion			Post: Q1-Q12			Post: Q13 onwards		
	Mean	SD	Count	Mean	SD	Count	Mean	SD	Count
D^N	0.153	0.588	12660	0.079	0.168	13883	0.042	0.107	21093
D^S	0.046	0.118	2355	0.137	0.304	9224	0.100	0.203	18776
β^{SI}	1.134	0.511	2424	1.230	0.611	8422	1.190	0.586	13413

Sustainable demand coordination ($D_{i,t}^S$) increases dramatically after inclusion, rising from 0.046 to 0.137 in the initial post-inclusion period, then stabilizing at 0.100 in the longer term. Conversely, conventional demand coordination ($D_{i,t}^N$) exhibits the opposite pattern, declining from 0.153 pre-inclusion to 0.079 initially post-inclusion, and further to 0.042 in the longer term. This divergence means conventional investors appear to reduce their coordinated response to these securities, potentially reflecting different information sets, investment horizons, or risk preferences.

This dramatic difference in observation counts illustrates the inclusion mechanism itself: sustainable funds rarely held event stocks before inclusion, generating few measurable coordination observations, while conventional funds maintained stable ownership patterns regardless of sustainable investing status. Figure 7 visualizes these dynamics around inclusion events, showing how coordination patterns evolve. The solid line demonstrates that sustainable investor demand coordination surges immediately after inclusion and stabilizes at an elevated level, consistent with the creation of a new systematic risk factor. Conversely, the dashed line shows conventional investor coordination declining steadily post-inclusion, indicating that conventional investors become less synchronized in their trading of SI stocks. The figure confirms that inclusion events successfully identify episodes where sustainable demand coordination emerges, providing the foundation for testing whether such coordination translates into beta changes.

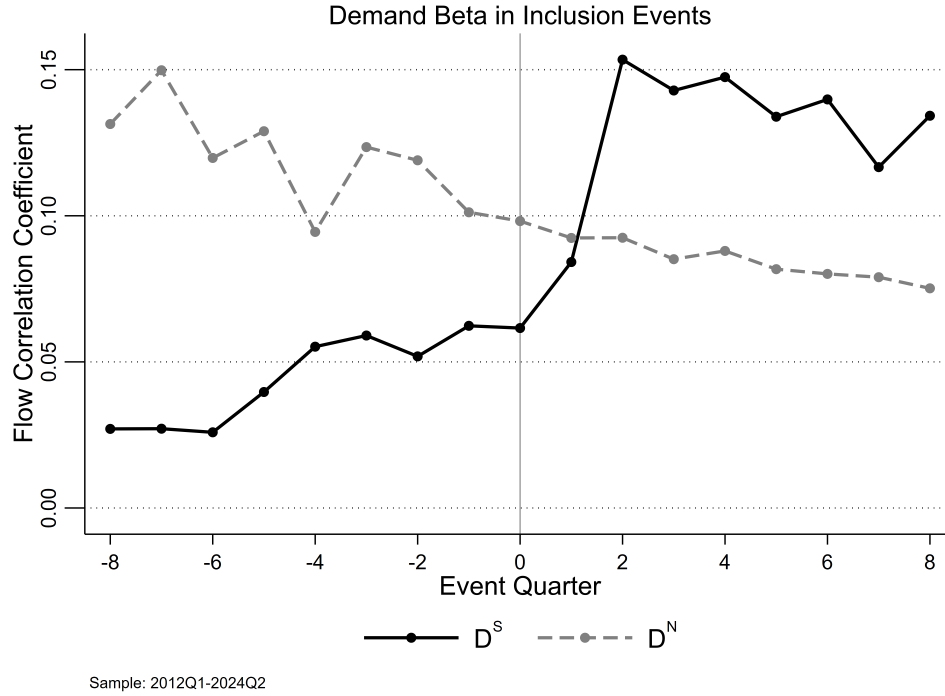


Figure 7. Demand Coefficient Dynamics for Event Stocks. This figure displays the average sustainable investor demand coefficient in solid and conventional investors in dash for event stocks around inclusion in the sustainable investing portfolio from 2012Q1 to 2024Q2. The demand coefficient measures the sensitivity of normalized flows into individual stocks to aggregate normalized flows into the SI portfolio, estimated using sustainable investing equity funds and conventional equity funds. I dropped the estimation where there were fewer than three fund owners of each type. The x-axis represents quarters relative to the inclusion quarter.

C.2. Baseline Regression

Using a three-year window before and after inclusion, I examine whether enhanced coordination translates into stronger price co-movement. Table VI presents regression results that provide strong support for the correlated demand channel. The coefficient of sustainable demand coordination $D_{i,t}^S$ increases dramatically from an insignificant -0.0008 pre-inclusion to a highly significant 0.120 (t-stat = 2.877) post-inclusion.

The economic magnitude is substantial: a one standard deviation increase in SI coordination (0.304 from the Q1-Q12 post inclusion) corresponds to a 0.036 increase in return beta (0.120×0.304). The full sample analysis confirms this relationship’s robustness, with the baseline specification yielding a coefficient of 0.116 (t-stat = 2.906). Even with controls and interaction terms, the core relationship remains significant at the 10% level, though the magnitude decreases slightly, suggesting some effect operates through stock characteristics.

The interaction terms provide insights into coordination heterogeneity. The negative coefficient on fund coverage interactions suggests stocks held by more SI funds experience diminishing marginal coordination effects, possibly reflecting diversification or capacity constraints.

C.3. Robustness and Other Analysis

To ensure the observed relationships reflect genuine SI coordination rather than spurious correlation, I conduct a placebo test using conventional fund flows. Table VII presents results that provide crucial validation of the coordinated demand mechanism.

The placebo test reveals strikingly different patterns from the main findings. Not only is there no statistically significant positive relationship between conventional fund coordination and return beta, but the controlled specification shows a significant negative coefficient (-0.078, t-stat = -2.110). This demonstrates that conventional investors do not contribute additional coordination to sustainable investing, as they actually react divergently between newly added stocks and the sustainable investing portfolio. Moreover, the regression coefficient is not statistically significant on average, indicating that conventional investor coordination is unlikely to drive cross-sectional variation in return betas.

This sharp contrast provides compelling evidence that the main results are not driven by generic flow-return relationships or mechanical correlations, but reflect coordination mecha-

Table VI
Sustainable Fund Flows Correlation and SI Portfolio Return Beta

This table shows the results of regressions of return betas β^{SI} with respect to the SI portfolio on coordinated SI demand D^S for all event stocks. Observations that were outside the interval of 12 quarters before and after the inclusion were excluded from this table.

$$\beta_{i,t}^{SI} = a_t + b^S D_{i,t}^S + C' X_{i,t} + \varepsilon_{i,t}$$

The dependent variable measures stock i 's return beta with the SI portfolio in each quarter t . The key explanatory variable $D_{i,t}^S$ captures sustainable fund coordination, estimated as the coefficient from equation (7), using sustainable investing fund flows. Controls include lagged market capitalization (*LagSize*), past year returns (*PastRet*), and interactions with the number of SI funds (*N_Funds*). All specifications include year-quarter fixed effects with standard errors clustered at the security level. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	β^{SI}			
	Pre-inclusion	Post-inclusion	Full Sample	Full sample
D^S	-0.000799 (-0.00715)	0.120*** (2.877)	0.116*** (2.906)	0.127* (1.815)
$N_Funds \times D^S$				-0.00471 (-0.412)
<i>LagSize</i>				-1.876*** (-3.215)
<i>PastRet</i>				-0.0304 (-0.972)
Constant	1.035*** (7.514)	1.036*** (14.17)	1.033*** (7.544)	1.183*** (11.48)
Year-Quarter FE	Yes	Yes	Yes	Yes
Observations	2,590	9,224	11,814	10,255
R-squared	0.040	0.065	0.058	0.062

nisms specific to sustainable investing.

Table VII
Conventional Fund Flow Correlation and SI Portfolio Return Beta

This table presents placebo test results examining the relationship between conventional fund demand coordination and SI portfolio return beta from 2012Q1 to 2024Q2. Observations that were outside the interval of 12 quarters before and after the inclusion were excluded from this table.

$$\beta_{i,t}^{SI} = a_t + b^N D_{i,t}^N + C' X_{i,t} + \varepsilon_{i,t}$$

The dependent variable is the return beta β_{it}^{SI} of security i with respect to the SI portfolio in quarter t . The main explanatory variable is the demand coordination $D_{i,t}^N$, which measures the sensitivity of normalized conventional fund flows into individual stocks to aggregate normalized conventional fund flows into the SI portfolio. Controls include interactions with the number of conventional fund owners, one-year lagged market capitalization, and one-year lagged returns. All specifications include year-quarter fixed effects with standard errors clustered at the security level. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	β^{SI}			
	Pre-inclusion	Post-inclusion	Full sample	Full sample
D^N	0.0234 (1.153)	0.0905 (1.314)	0.0304 (1.188)	-0.0780** (-2.110)
$N_Funds \times D^N$				0.0102*** (3.813)
$LagSize$				-2.318*** (-3.960)
$PastRet$				0.0933*** (3.896)
Constant	1.343*** (41.61)	1.184*** (23.86)	1.274*** (51.49)	1.271*** (50.39)
Year-Quarter FE	Yes	Yes	Yes	Yes
R-squared	0.064	0.073	0.069	0.078
Observations	8,781	13,883	21,787	21,787

Table VIII presents the results of bivariate regressions that include both SI and conventional funds demand coefficients, providing a direct comparison of their relative importance in explaining return beta dynamics. I merge the two demand coefficients into a whole panel of data. When a stock lacks sufficient ownership by either fund type (fewer than three funds), I set the corresponding coordination measure to zero, reflecting the absence of coordinated trading by that investor group. This approach provides a conservative test of relative importance while maintaining the full sample.

The joint analysis reinforces the primacy of sustainable investor demand coordination. In the third column, sustainable demand has a positive coefficient nearly 4 times larger than conventional demand (0.0989 vs. 0.0259), and only sustainable demand shows a growing impact on return beta over time, while conventional demand remains relatively stable.

The last column includes stock size and past performance as controls. Consistent with traditional asset pricing papers, small firms and past winners exhibit higher risk exposure. However, even after controlling for these traditional factors, sustainable investor demand coordination maintains its positive and significant effect (coefficient = 0.116, t-stat = 2.414), while the conventional investor demand coefficient becomes significantly negative (-0.086, t-stat = -2.214). This pattern demonstrates that these represent distinct and opposing coordination mechanisms, with sustainable investor demand coordination driving systematic risk creation while conventional investor demand coordination appears to have a dampening effect.

I further examine the timing of these relationships around inclusion events. If coordinated demand drives price correlation, demand coordination measured around the inclusion period should predict systematic risk changes in subsequent years rather than vice versa. I test this prediction by regressing systematic risk exposure in post-inclusion years on demand coordination measures from 2 quarters before to 2 quarters after inclusion ($t - 2, t - 1, t, t + 1, t + 2$).

Table A4 reveals evolving patterns over time. In the first year post-inclusion, preceding demand coordination ($D_{i,t-1}^S$) significantly predicts systematic risk exposure, consistent with demand-driven co-movement. However, by the third year, forward-looking demand coordination ($D_{i,t+1}^S$) becomes significant, suggesting that systematic risk exposure begins to influence subsequent demand patterns.

V. Conclusion

This paper documents the price movements in sustainable investing and identifies a novel correlated demand mechanism that engenders beta changes in ESG-focused portfolios. Based on the analysis of the portfolio allocations from sustainable active investing mutual funds between 2012 and 2024, this paper demonstrates that stocks experience significant increases in return beta after being exposed to the sustainable investors' demand. First, stocks pre-

Table VIII
Bivariate Regression

This table presents bivariate regressions examining both SI and conventional fund coordination simultaneously from 2012Q1 to 2024Q2. Observations that were outside the interval of 12 quarters before and after the inclusion were excluded from this table. The dependent variable $\beta_{i,t}^{SI}$ measures stock i 's return beta with the SI portfolio. Key explanatory variables are $D_{i,t}^S$ (SI fund coordination) and $D_{i,t}^N$ (conventional fund coordination), both estimated using the same methodology but applied to different fund groups. When stocks have insufficient ownership by either fund type, the corresponding coordination measure is set to zero. Controls include interactions with fund coverage (the number of each type of fund owner), lagged market capitalization, and past returns. All specifications include year-quarter fixed effects with standard errors clustered at the security level. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	β^{SI}			
	Pre-inclusion	Post-inclusion	Full sample	Full sample
D^S	-0.0531 (-0.397)	0.101*** (2.595)	0.0989*** (2.640)	0.116** (2.414)
D^N	0.0177 (0.941)	0.0800 (1.174)	0.0259 (1.036)	-0.0861** (-2.214)
$D^S \times N_funds^S$				-0.00780 (-0.829)
$D^N \times N_funds^N$				0.0106*** (3.894)
$LagSize$				-2.172*** (-3.726)
$PastRet$				0.0904*** (3.804)
Constant	1.362*** (37.49)	1.184*** (23.86)	1.275*** (51.14)	1.271*** (49.90)
Year-Quarter FE	Yes	Yes	Yes	Yes
Observations	7,637	13,928	21,441	21,441
R-squared	0.068	0.075	0.070	0.078

ferred in the SI universe show concentrated co-movement during the 2016-2020 period; on average, one-year return betas increased 7% after stocks were overweighted by sustainable investors, relative to matched control stocks. Second, the traditional fundamental factor models lose explanatory power for these stocks, suggesting that fundamental factor loading changes cannot fully account for the observed risk exposure shifts. Third, the correlated demand mechanism provides a compelling explanation for these patterns. In particular, the coordinated demand of sustainable investing mutual funds channels a new source of systematic risk into the stocks overweighted by sustainable funds. The identification relies on ESG-focused coordination being distinct from broader market coordination patterns, which the differential response between sustainable and conventional investors supports. For the same event stock, sustainable investors exhibit strong coordination patterns, while conventional investors show no such coordination for identical stocks during the same time periods. This asymmetric response pattern strengthens identification by ruling out explanations based on generic stock characteristics or time-varying market factors that would affect all investor types equally. Overall, this paper shows how demand-driven risk affects the sustainable investment price dynamics. It suggests that market structure evolution and investor behavior adaptation continue to shape the sustainable investing landscape.

REFERENCES

- Albuquerque, Rui, Yrjö Koskinen, and Chendi Zhang, 2019, Corporate social responsibility and firm risk: Theory and empirical evidence, *Management science* 65, 4451–4469.
- Anton, Miguel and Christopher Polk, 2014, Connected stocks, *The Journal of Finance* 69, 1099–1127.
- Avramov, Doron et al., 2022, Sustainable investing with ESG rating uncertainty, *Journal of financial economics* 145, 642–664.
- Barberis, Nicholas and Andrei Shleifer, 2003, Style investing, *Journal of financial Economics* 68, 161–199.
- Barberis, Nicholas, Andrei Shleifer, and Jeffrey Wurgler, 2005, Comovement, *Journal of financial economics* 75, 283–317.
- Basak, Suleyman and Anna Pavlova, 2013, Asset prices and institutional investors, *American Economic Review* 103, 1728–1758.
- Birk, Kevin, Stefan Jacob, and Marco Wilkens, 2024, What attracts sustainable fund flows? Prospectus versus ratings, *Journal of Asset Management*, 1–22.
- Cao, Jie et al., 2023, ESG preference, institutional trading, and stock return patterns, *Journal of financial and quantitative analysis* 58, 1843–1877.
- Ceccarelli, Marco et al., 2023, ESG skill of mutual fund managers.
- Chen, Honghui, Vijay Singal, and Robert F Whitelaw, 2016, Comovement revisited, *Journal of Financial Economics* 121, 624–644.
- Ciciretti, Rocco, Ambrogio Dalò, and Giovanni Ferri, 2021, Herding and anti-herding across ESG funds.
- Dorn, Daniel, Gur Huberman, and Paul Sengmueller, 2008, Correlated trading and returns, *The Journal of Finance* 63, 885–920.
- Dumitrescu, Ariadna and Javier Gil-Bazo, 2024, ESG and Mutual Fund Competition, *Available at SSRN 4887875*.
- Edmans, Alex, Tom Gosling, and Dirk Jenter, 2024, Sustainable investing: Evidence from the field, *FEB-RN Research Paper*.
- Geczy, Christopher C, Robert F Stambaugh, and David Levin, 2021, Investing in socially responsible mutual funds, *The Review of Asset Pricing Studies* 11, 309–351.
- Giese, Guido et al., 2019, Foundations of ESG investing: How ESG affects equity valuation, risk, and performance, *Journal of portfolio management* 45, 69–83.

- Giglio, Stefano et al., 2025, Four facts about ESG beliefs and investor portfolios, *Journal of Financial Economics* 164, 103984.
- Green, Daniel and Benjamin N Roth, 2021, The allocation of socially responsible capital, *The Journal of Finance*.
- Green and Hwang, 2009, Price-based return comovement, *Journal of Financial Economics* 93, 37–50.
- Green, T Clifton and Byoung-Hyoun Hwang, 2009, Price-based return comovement, *Journal of financial economics* 93, 37–50.
- Greenwood, Robin and David Thesmar, 2011, Stock price fragility, *Journal of Financial Economics* 102, 471–490.
- Grégoire, Vincent, 2020, The rise of passive investing and index-linked comovement, *The North American Journal of Economics and Finance* 51, 101059.
- Han, Xiao, Nikolai L Roussanov, and Hongxun Ruan, 2021, Mutual fund risk shifting and risk anomalies, *Jacobs Levy Equity Management Center for Quantitative Financial Research Paper*.
- Jeffers, Jessica, Tianshu Lyu, and Kelly Posenau, 2024, The risk and return of impact investing funds, *Journal of Financial Economics* 161, 103928.
- Kasch, Maria and Asani Sarkar, 2013, Is there an S&P 500 index effect?
- Kim, Byungwook, 2025, Correlated Demand Shocks and Asset Pricing, *Available at SSRN 4705500*.
- Koijen, Ralph SJ, Robert J Richmond, and Motohiro Yogo, 2023, Which investors matter for equity valuations and expected returns?, *Review of Economic Studies*, rdad083.
- Martí-Ballester, Carmen Pilar, 2015, Investor reactions to socially responsible investment, *Management Decision* 53, 571–604.
- Noh, Don, Sangmin S Oh, and Jihong Song, 2023, Unpacking the demand for sustainable equity investing, *SSRN Electron. J.*
- Pastor, Lubos, Robert F Stambaugh, and Lucian A Taylor (2024). *Sustainable Investing*. Tech. rep. National Bureau of Economic Research.
- Pástor, L’uboš, Robert F Stambaugh, and Lucian A Taylor, 2021, Sustainable investing in equilibrium, *Journal of financial economics* 142, 550–571.
- Pástor, L’uboš, Robert F Stambaugh, and Lucian A Taylor, 2022, Dissecting green returns, *Journal of financial economics* 146, 403–424.

- Pedersen, Lasse Heje, Shaun Fitzgibbons, and Lukasz Pomorski, 2021, Responsible investing: The ESG-efficient frontier, *Journal of financial economics* 142, 572–597.
- Rzeźnik, Aleksandra, Kathleen Weiss Hanley, and Loriana Pelizzon, 2022, Investor reliance on ESG ratings and stock price performance.
- Starks, Laura T, 2023, Presidential address: Sustainable finance and esg issues—value versus values, *The Journal of Finance* 78, 1837–1872.
- Van der Beck, Philippe, 2021, Flow-driven ESG returns, *Swiss Finance Institute Research Paper*.
- Wahal, Sunil and Albert Yan Wang, 2011, Competition among mutual funds, *Journal of Financial Economics* 99, 40–59.

Appendix

A. Tables and Figures

Table A1
Sustainable Investment Universe Growth

This table lists the number of all stocks in the SI Universe from 2012 Q1 to 2024 Q2. SI Stocks are those where portfolio weight exceeds market portfolio weight, indicating active overweighting by sustainable funds. The S&P 500 column shows newly added S&P 500 constituents each quarter. The column of "New" shows the number of newly added stocks in each expansion.

Quarter	SI Universe		SI Stocks		S&P 500
	New	Total	New	Total	
2012q1	381	788	129	411	168
2012q2	38	826	47	458	59
2012q3	12	838	37	495	16
2012q4	15	853	31	526	14
2013q1	57	910	86	612	11
2013q2	34	944	10	622	31
2013q3	17	961	33	655	4
2013q4	19	980	23	678	14
2014q1	28	1008	26	704	7
2014q2	45	1053	46	750	7
2014q3	21	1074	23	773	13
2014q4	34	1108	16	789	8
2015q1	53	1161	38	827	4
2015q2	40	1201	14	841	3
2015q3	21	1222	23	864	4
2015q4	24	1246	35	899	11
2016q1	28	1274	68	967	10
2016q2	31	1305	20	987	17
2016q3	17	1322	22	1009	4
2016q4	30	1352	27	1036	5
2017q1	32	1384	54	1090	6
2017q2	140	1524	23	1113	9
2017q3	29	1553	22	1135	4
2017q4	47	1600	19	1154	5
2018q1	45	1645	32	1186	2
2018q2	43	1688	23	1209	17
2018q3	26	1714	12	1221	2
2018q4	33	1747	24	1245	2
2019q1	60	1807	65	1310	5
2019q2	34	1841	16	1326	13
2019q3	55	1896	19	1345	3
2019q4	36	1932	14	1359	5
2020q1	145	2077	99	1458	13
2020q2	40	2117	39	1497	3
2020q3	40	2157	33	1530	5
2020q4	42	2199	29	1559	3
2021q1	62	2261	65	1624	8
2021q2	90	2351	44	1668	2
2021q3	65	2416	29	1697	2
2021q4	53	2469	44	1741	0
2022q1	54	2523	65	1806	6
2022q2	64	2587	18	1824	0
2022q3	35	2622	15	1839	0
2022q4	35	2657	13	1852	1
2023q1	116	2773	28	1880	1
2023q2	139	2912	43	1923	3
2023q3	43	2955	24	1947	3
2023q4	57	3012	27	1974	3
2024q1	71	3083	25	1999	1
2024q2	136	3219	28	2027	1

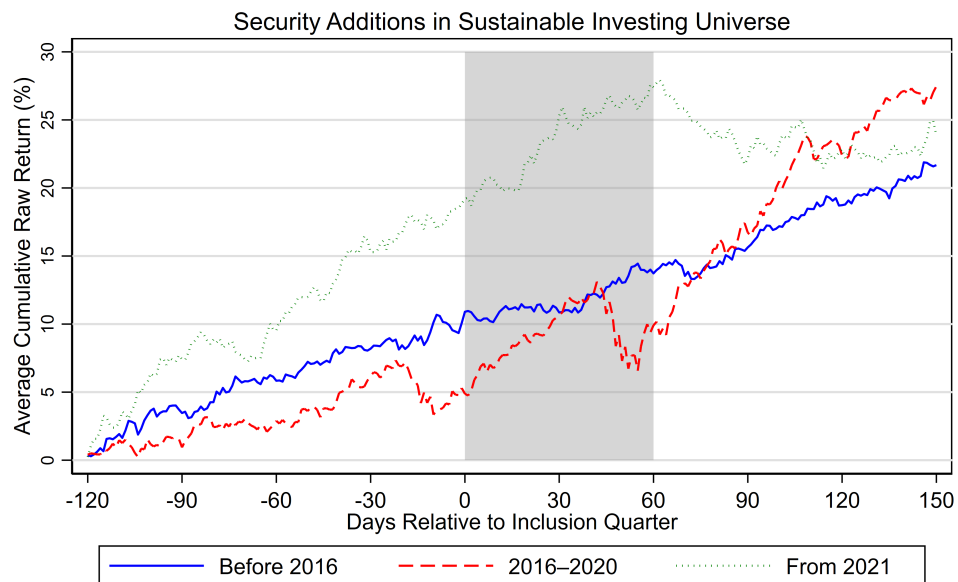


Figure A1. Cumulative Raw Returns of Event Stocks. The figure presents the cumulative raw returns for inclusion events that occurred during different periods: before 2016 (blue line), between 2016 and 2020 (red line), and 2021 onward (green line). The shaded gray area indicates the inclusion quarter. The analysis spans 120 days before the first day of the event quarter through 150 days after.

Table A2
Change in Market Beta by Period and Frequency

This table presents changes in market beta at daily, weekly, and monthly frequencies. The market beta is estimated as:

$$\beta_t^{Mkt} = \frac{\text{cov}_t(r_{i,t}, R_t^{Mkt})}{\text{var}_t(R_t^{Mkt})}$$

where R_t^{Mkt} is the CRSP value-weighted market portfolio return and $r_{i,t}$ is the return of stock i at time t . $\Delta\beta = \beta_{post} - \beta_{pre}$ measures the difference between pre- and post-inclusion market betas. For daily and weekly frequencies (Panels A and B), estimation periods span 12 months before and after the inclusion quarter: $[-12, -1]$ and $[1, 12]$. For the monthly frequency (Panel C), estimation periods expand to 36 months. The sample is divided into three sub-periods: 2013–2015, 2016–2019, and 2020–2023. Stocks with less than 80% of available trading days during the estimation window are excluded. Standard errors are reported in the final column.

Panel A: Daily Market Beta			
Period	Observations	$\Delta\beta^{Mkt}$	Std Err
2013–2015	336	-0.0329	0.0180
2016–2019	375	0.0815	0.0255
2020–2023	480	0.0403	0.0243
Panel C: Weekly Market Beta			
Period	Observations	$\Delta\beta^{Mkt}$	Std Err
2013–2015	336	-0.0260	0.0303
2016–2019	372	0.0860	0.0426
2020–2023	477	0.0062	0.0420
Panel E: Monthly Market Beta			
Period	Observations	$\Delta\beta^{Mkt}$	Std Err
2013–2015	304	-0.0571	0.0384
2016–2019	305	0.1185	0.0477
After 2020	351	-0.0870	0.0406

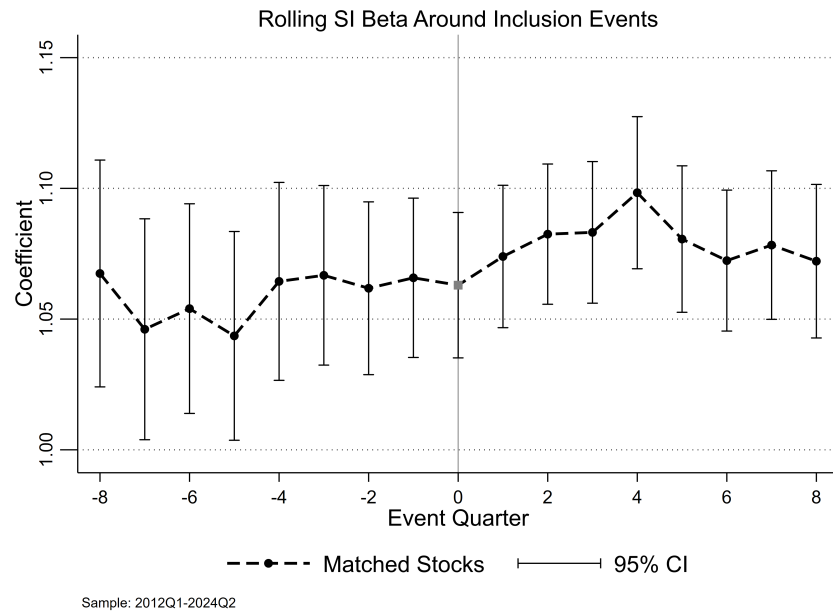
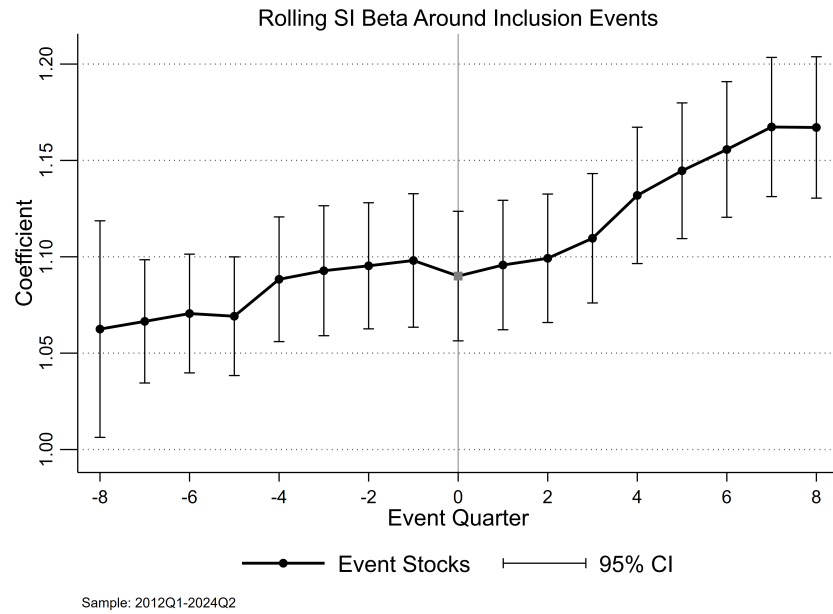


Figure A2. Rolling Beta on SI portfolio of Event stocks and Matched Stocks
 This figure presents the one-year rolling beta coefficients for event stocks (solid line) and their matched control stocks (dashed line) relative to the SI portfolio with 95% confidence interval around the beta estimates. Rolling betas are calculated using daily returns over 252 trading days and updated quarterly.

Table A3
Correlation and Volatility Changes Among Event Stocks

This table examines changes in correlation and volatility among event stocks following inclusion in the sustainable investing universe from 2012Q1 to 2024Q2. Δ Correlation measures the change in correlation coefficients between individual event stocks and their peer portfolios computed using 12-month daily returns before and after the event quarter. Stocks with less than 80% of available trading days during the estimation window were excluded. Δ Individual Volatility measures the change in standard deviation of 12-month daily returns for individual event stocks, while Δ Portfolio Volatility measures the change in volatility of peer portfolios constructed from other event stocks sharing the same event quarter. Observations with less than one year of trading data were excluded from this table. Panel A analyzes the period before 2016, Panel B covers 2016–2020, Panel C examines the period from 2021. The analysis tests whether correlation and volatility among event stocks change systematically after inclusion, indicating enhanced co-movement and new risk dynamics driven by correlated demand within the sustainable investing ecosystem. Robust t -statistics are reported in parentheses. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Before 2016			
	Δ Correlation	Δ Individual Volatility	Δ Portfolio Volatility
	-0.0213*** (-2.961)	-0.000748** (-2.373)	-0.000481*** (-2.981)
Observations	331	331	331
Panel B: 2016–2020			
	Δ Correlation	Δ Individual Volatility	Δ Portfolio Volatility
	0.0164* (1.963)	0.00245*** (3.982)	0.00174*** (5.656)
Observations	511	511	511
Panel C: From 2021			
	Δ Correlation	Δ Individual Volatility	Δ Portfolio Volatility
	-0.00205 (-0.173)	-0.00236** (-2.440)	-0.000945*** (-2.788)
Observations	229	229	229

Table A4
Timing of Demand Coordination and Return Beta Changes

This table examines the causal relationship between demand coordination and systematic risk exposure by analyzing the timing of these relationships around inclusion events in the sustainable investing universe. The regression equation is:

$$\beta_{i,t}^{SI} = \rho + \sum_{n=-2}^2 \phi_n D_{i,t+n}^{(S)} + a_t + w_{i,t}$$

where $\beta_{i,t}^{SI}$ is the systematic risk exposure of stock i in quarter t relative to inclusion, and $D_{i,t+n}^{(S)}$ represents the demand coordination measure for stock i at time $t+n$. The demand coordination measure is defined in Eq. 7 Column (1) estimates the influence of demand coordination on systematic risk in the first year post-inclusion [quarters 0-4]. Columns (2) and (3) present estimates for the second [quarters 4-8] and third years [quarters 8-12], respectively. If coordinated demand drives price correlation, lagged demand ($D_{i,t-1}^{(S)}$, $D_{i,t-2}^{(S)}$) should predict subsequent return beta changes. Demand beta is set to zero when no sustainable investor trades the stock. All regressions include time fixed effects. Robust t -statistics are reported in parentheses, clustering at the firm level. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Post-inclusion	β^{SI}		
	Year 1	Year 2	Year 3
$D_{i,t}^{(S)}$	0.0199 (0.376)	0.147*** (3.434)	0.0885 (1.338)
$D_{i,t-1}^{(S)}$	0.115** (2.118)	-0.00905 (-0.318)	0.0548 (1.308)
$D_{i,t-2}^{(S)}$	0.0649 (1.031)	0.148*** (2.864)	-0.0158 (-0.456)
$D_{i,t+1}^{(S)}$	0.00272 (0.0581)	0.00839 (0.257)	0.177*** (3.423)
$D_{i,t+2}^{(S)}$	-0.0236 (-0.663)	-0.0204 (-0.389)	0.0236 (0.340)
Constant	1.196*** (24.56)	1.088*** (33.55)	1.088*** (20.15)
Time FE	Yes	Yes	Yes
Observations	5,624	5,355	4,958
Number of Stocks	1,406	1,367	1,302
R-squared	0.062	0.061	0.097

B. Measuring Correlated Demand

This section develops the theoretical foundation for measuring correlated fund flows following Greenwood and Thesmar 2011. The framework links portfolio weight changes to price movements in a closed economy where all investor holdings are observable.

Portfolio Weight Decomposition

For representative investor k , the dollar weight of security i at time t is:

$$w_{ikt} = \frac{n_{ikt}P_{it}}{A_{kt}} \quad (9)$$

where n_{ikt} is the number of shares held, P_{it} is the price, and A_{kt} is total portfolio value.

Through log-linearization, net purchases can be decomposed as:

$$P_{it}\Delta n_{ikt} = P_{it}n_{ikt} \left(\frac{\Delta w_{ikt}}{w_{ikt}} - \frac{\Delta P_{it}}{P_{it}} \right) + P_{it}w_{ikt}\Delta A_{kt} \quad (10)$$

Portfolio asset changes consist of net inflows and capital gains:

$$\Delta A_{kt} = f_{kt} + \sum_j \text{Cap}_{jkt} \quad (11)$$

Active Trading versus Flow-Induced Trading

Dividing by P_{it} , net share changes decompose into two components:

$$\Delta n_{ikt} = \underbrace{n_{ikt} \left(\frac{\Delta w_{ikt}}{w_{ikt}} - \frac{\Delta P_{it}}{P_{it}} \right)}_{\text{Active Rebalancing}} + \underbrace{w_{ikt} \left(\frac{f_{kt}}{P_{it}} + \frac{\sum_j \text{Cap}_{jkt}}{P_{it}} \right)}_{\text{Flow-Induced Trading}} \quad (12)$$

Active rebalancing reflects deliberate portfolio weight adjustments, while flow-induced trading results from inflows and capital gains distributed proportionally.

Equilibrium Price Determination

Aggregating across all investors and assuming fixed supply ($\sum_k \Delta n_{ikt} = 0$), equilibrium price changes are:

$$\frac{\Delta P_{it}}{P_{it}} = \frac{\sum_k n_{ikt} \frac{\Delta w_{ikt}}{w_{ikt}}}{\sum_k n_{ikt}} + \frac{\sum_k w_{ikt} \left(\frac{f_{kt}}{P_{it}} + \frac{\sum_j \text{Cap}_{jkt}}{P_{it}} \right)}{\sum_k n_{ikt}} \quad (13)$$

This links price movements to aggregate portfolio adjustments and fund flows.

Flow Beta Construction

The framework implies that asset returns depend on aggregate flow-driven demand:

$$r_{it+1} = \psi_{it} + \lambda_i \frac{\sum_k w_{ikt} f_{kt}}{\theta_{it}} + \epsilon_{it+1} \quad (14)$$

where θ_{it} is a scaling factor (typically market capitalization) and ϵ_{it+1} represents fundamental news.

The correlation between asset prices, therefore, reflects the correlation between capital flows. For sustainable investing, this motivates measuring flow beta as the covariance between individual stock flows and portfolio flows:

$$F_{it} = \frac{\text{cov}_t \left(\frac{\sum_k w_{ikt} f_{kt}}{\theta_{it}}, \sum_j s_{jt} \frac{\sum_k w_{jkt} f_{kt}}{\theta_{jt}} \right)}{\text{var}_t \left(\sum_j s_{jt} \frac{\sum_k w_{jkt} f_{kt}}{\theta_{jt}} \right)} \quad (15)$$

This flow beta measures how individual stock flows correlate with aggregate portfolio flows, where the portfolio is a value-weighted average of all current members in the sustainable investing universe. The portfolio weight $s_{jt} = \theta_{jt} / \sum_j \theta_{jt}$ is based on each stock's market capitalization θ_{jt} . This measure captures what T. C. Green and B.-H. Hwang (2009) term "co-fragility", the mechanism through which correlated demand translates into systematic return comovement.

Derivation of the Relation Between $D_{i,t}$ and $F_{i,t}$

Starting from price fragility measures between stock i and SI portfolio, write (15) as

$$F_{it} = \frac{\text{cov}_t\left(\sum_k \frac{w_{ikt}f_{kt}}{\theta_{it}}, \sum_{j \in SI} \frac{\sum_k w_{jkt}f_{kt}}{\sum_j \theta_{jt}}\right)}{\text{var}_t\left(\sum_{j \in SI} \frac{\sum_k w_{jkt}f_{kt}}{\sum_j \theta_{jt}}\right)} \quad (16)$$

Using the cross-sectional regression estimates from (7), for each i and at the end of t , we have:

$$\sum_k \frac{w_{ikt}f_{kt}}{\theta_{it}} = \sum_s \frac{w_{ist}f_{st}}{\theta_{it}} + \sum_n \frac{w_{int}f_{nt}}{\theta_{it}} \quad (17)$$

$$\approx N^{SI} D_{i,t}^S \frac{\sum_j w_{jst}f_{st}}{\sum_j \theta_{jt}} + N^{NonSI} D_{i,t}^N \frac{\sum_j w_{jnt}f_{nt}}{\sum_j \theta_{jt}} \quad (18)$$

where $D_{i,t}^S$ and $D_{i,t}^N$ are the cross-sectional coefficients across sustainable and conventional funds, respectively. k is the notation for fund regardless of fund type. s represents the sustainable funds, and n indicates the conventional funds. N^{SI} and N^{NonSI} are the number of sustainable and conventional funds at time t . Thus, the total fund flows are approximated as a linear combination of contributions from SI capital flows and conventional capital flows, weighted by the number of investors.

Substituting into the covariance, we get

$$\begin{aligned} \text{cov}_t\left(\sum_k \frac{w_{ikt}f_{kt}}{\theta_{it}}, \sum_j \frac{\sum_k w_{jkt}f_{kt}}{\sum_j \theta_{jt}}\right) &\approx N^{SI} D_{i,t}^S \text{cov}_t\left(\sum_j \frac{\sum_s w_{jst}f_{st}}{\sum_j \theta_{jt}}, \sum_j \frac{\sum_k w_{jkt}f_{kt}}{\sum_j \theta_{jt}}\right) \\ &+ N^{NonSI} D_{i,t}^N \text{cov}_t\left(\sum_j \frac{\sum_n w_{jnt}f_{nt}}{\sum_j \theta_{jt}}, \sum_j \frac{\sum_k w_{jkt}f_{kt}}{\sum_j \theta_{jt}}\right). \end{aligned}$$

Dividing by the variance of total SI flows, we obtain the compact relation:

$$\begin{aligned} F_{i,t} &\approx N^{SI} D_{i,t}^S V_t^{SI} + N^{NonSI} D_{i,t}^N V_t^{NonSI} \\ V_t^{SI} &= \frac{\text{cov}_t(\text{SI Fund flows, Total flows})}{\text{var}_t(\text{Total flows})} \end{aligned}$$

$$V_t^{NonSI} = \frac{\text{cov}_t(\text{NonSI Fund flows, Total flows})}{\text{var}_t(\text{Total flows})}$$

Interpretation: $D_{i,t}^S$ and $D_{i,t}^N$ are estimated from the cross-section at time t (across k), which capture the cross-sectional variations in stock prices that are driven by the heterogeneous demand shocks from different funds k . In contrast, $F_{i,t}$ incorporates the time-series covariance structure of capital flows between stock i and aggregate flows in the SI portfolio, containing both cross-sectional exposure and temporal capital flow volatility. $F_{i,t}$ can be dismantled into the contributions from different fund types, as shown in (18). The first term is the contribution from the SI fund flows-induced trading. This is typically the dominant term for funds heavily invested in SI stocks. The second term captures the contribution from NonSI fund flows. It is crucial that the decomposition allows us to isolate the effects of targeted SI demand from broader market movements. Overall, by linking the price co-movements with D^S and D^N while controlling for the time-fixed effects, it shows transparently how variation in investor demand across securities drives stock prices distinctively, which helps detect the subgroup risk related to unique investor demand.

C. Notations, Definitions, and Sample Construction

Table C1
Portfolio and Return Measures

Notation	Definition	Explanation
R_t^{SI}	SI Portfolio Return	$R_t^{SI} = \sum_j s_{j,t}^{SI} r_{j,t}$ where $s_{j,t}^{SI}$ is the weight of stock j
$s_{j,t}^{SI}$	SI Portfolio Weight	$s_{j,t}^{SI} = \frac{\theta_{j,t}}{\sum_{j \in SI} \theta_{j,t}}$, market-cap weighted portfolio share
$r_{i,t}$	Individual Stock Return	Return of stock i from period t to $t + 1$
R_t^{CRSP}	CRSP Market Return	Value-weighted (VW) or equally-weighted (EW) CRSP market index return
R_t^{Non-SI}	Non-SI Portfolio Return	Size-adjusted return of non-sustainable securities
$\theta_{i,t}$	Market Capitalization	Market value of stock i at time t (scaling factor)

Table C2
Beta Coefficients and Risk Measures

Notation	Definition	Explanation
β_i^{SI}	SI Portfolio Beta	$\beta_i^{SI} = \frac{\text{cov}_t(r_{i,t}, R_t^{SI})}{\text{var}_t(R_t^{SI})}$, stock's sensitivity to SI portfolio
β_i^p	General Portfolio Beta	Stock i 's sensitivity to portfolio p returns
$\Delta\beta$	Beta Change	$\Delta\beta = \beta_{post} - \beta_{pre}$, change in beta around inclusion
$\Delta\Delta\beta$	Difference-in-Differences Beta	Excess beta change of event stocks vs. matched controls
$\eta_{i,t}$	Fundamental Loading	Factor Stock i 's exposure to fundamental risk factor f_t
$\gamma_{i,t}$	SI-Specific Risk Loading	Stock i 's exposure to SI-specific group risk μ_t^{SI}

Table C3
Fund Flow and Demand Variables

Notation	Definition	Explanation
$f_{k,t}$	Fund Flow	Dollar flows into fund k from period t to $t + 1$
$w_{i,k,t}$	Portfolio Weight	Weight of stock i in fund k 's portfolio at time t
$D_{i,t}^S$	Sustainable Fund Demand Coordination	Coordination coefficient for sustainable funds' flows into stock i
$D_{i,t}^N$	Conventional Fund Demand Coordination	Coordination coefficient for conventional funds' flows into stock i
$F_{i,t}$	Co-fragility	Flow-based beta measuring correlated demand exposure
$\frac{w_{i,k,t} f_{k,t}}{\theta_{i,t}}$	Normalized Flow	Fund flow into stock i normalized by market capitalization

Table C4
Risk Factors and Model Components

Notation	Definition	Explanation
f_t	Fundamental Factor	Shared fundamental risk factor
μ_t^{SI}	SI Group Risk	SI-specific group risk factor, $\mu_t^{SI} \sim N(0, \sigma_\mu^2)$
$e_{i,t}$	Idiosyncratic Risk	Stock-specific risk, $e_{i,t} \sim N(0, \sigma_e^2)$
$\epsilon_{i,t+1}$	Return Error Term	Residual return component with $E[\epsilon_{i,t+1}] = 0$
$u_{i,t}$	Flow Regression Error	Error term in demand beta regressions
α_i	Intercept	Regression intercept for stock i
λ_i	Flow Sensitivity	Price sensitivity to normalized flows for stock i
a_t	Time Fixed Effect	Quarter-specific fixed effect

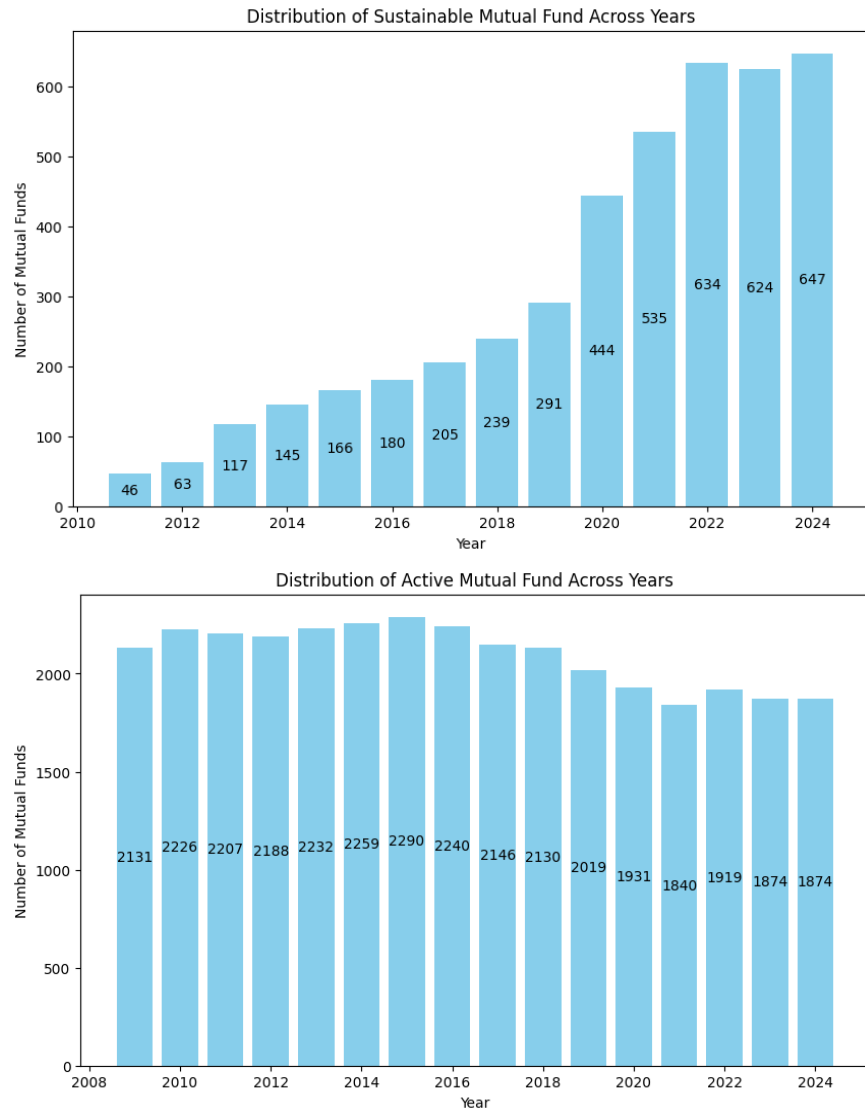


Figure C1. Growth of Sustainable Mutual Funds in U.S. Market, 2011-2024 This figure presents the number of sustainable mutual funds and all active equity mutual funds in the sample period. This figure excludes all dead funds or funds that have merged into another in each fiscal year.

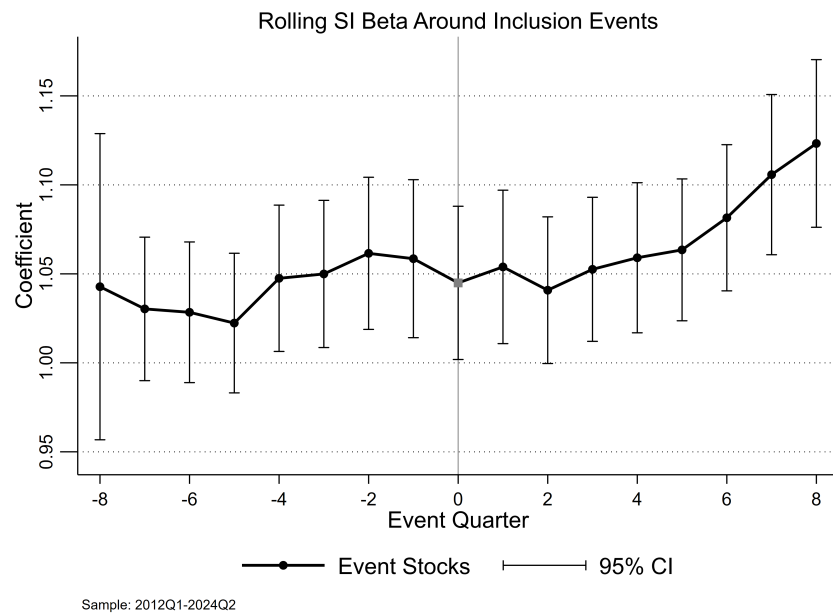
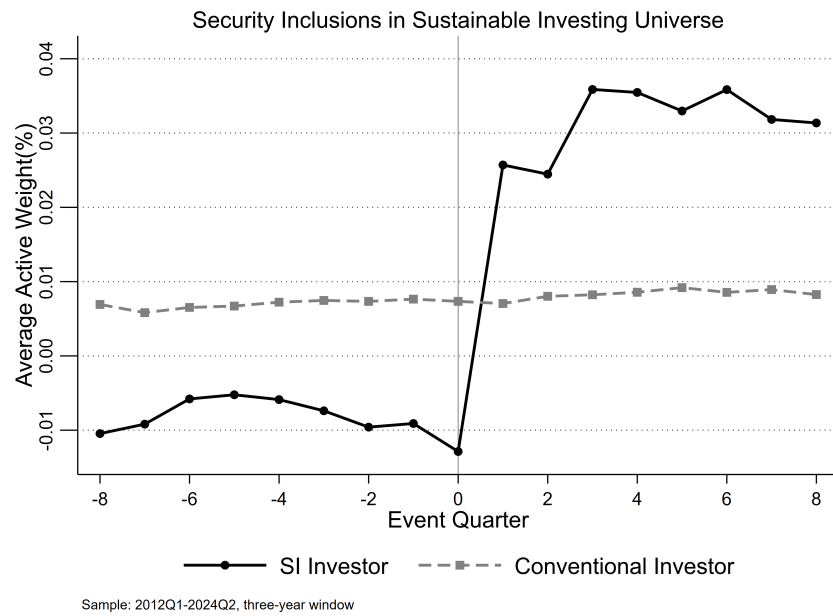


Figure C2. Portfolio Weight Changes and Rolling Beta for Alternative Event Definition This figure presents results using the alternative identification of event stocks requiring 12 consecutive positive active weights. The black line shows active positions by sustainable investors, where portfolio weights in the SI universe exceed market portfolio weights. The gray dashed line represents conventional investors, defined as US active equity mutual funds that do not follow specific ESG investment strategies. This alternative definition yields 874 event stocks compared to 1,406 in the baseline specification. Rolling betas around inclusion events are calculated using daily returns over 252 trading days and updated quarterly.

Table C5
Control Variables and Sample Definitions

Notation	Definition	Explanation
$X_{i,t}$	Control Vector	Vector of time-varying stock characteristics
$Size$	Market Capitalization	Stock market capitalization (often in billions or deciles)
$PastRet$	Past Returns	Cumulative returns over past 4 quarters or 1 year
N_Funds	Fund Coverage	Number of funds holding the stock
CAP_t^{SI}	SI Market Cap	Total market capitalization of SI universe
CAP_t^{CRSP}	CRSP Market Cap	Total CRSP market capitalization
b	Main Coefficient	Coefficient on SI demand beta in regression
C	Control Coefficients	Vector of coefficients on control variables

Table C6
Sample Definitions and Time Notation

Notation/Term	Definition	Explanation
Sample Definitions		
SI Universe	Sustainable Investing Universe	All stocks held by at least one sustainable fund
Event Stock	Inclusion Event Stock	Stock newly overweighted in SI portfolio for ≥ 12 quarters
SI Stock	Overweight SI Stock	Stock with positive active weight in SI universe
Active Weight	Relative Weight	Deviation of SI weight from market portfolio weight
Pre/Post	Event Periods	12 months before/after inclusion quarter
Time Notation		
t	Time Period	Usually refers to quarters (Q1, Q2, Q3, Q4)
$t - 1, t + 1$	Lagged/Lead Periods	Previous and next periods relative to t
$[-12, -1]$	Pre-Event Window	12 months before inclusion quarter
$[1, 12]$	Post-Event Window	12 months after inclusion quarter
2012Q1 – 2024Q2	Sample Period	Full sample period from 2012 Q1 to 2024 Q2